

Highlighted Manuscript: Change and Deletion Index

Baseline: the manuscript as submitted through the IEEE Access portal. Yellow marks show inserted/replaced rendered words; orange frames identify pages containing changed figure assets. The clean PDF text is unchanged by annotations. Source-only layout/preamble edits are preserved in reviewed-baseline-source.diff. This index records removed text that cannot be highlighted in the revised manuscript.

Change 1: replace | old pp. [1] | new pp. [1]

Before (including deletions): SPICE

After: Structural Circuit

Change 2: replace | old pp. [1] | new pp. [1]

Before (including deletions): Automated digitization of

After: Recovering a netlist from a

Change 3: replace | old pp. [1] | new pp. [1]

Before (including deletions): circuit schematics into SPICE netlists remains challenging because connectivity recovery is brittle under noisy

After: schematic requires deciding which

Change 4: replace | old pp. [1] | new pp. [1]

Before (including deletions): detections.

After: strokes connect which component terminals.

Change 5: replace | old pp. [1] | new pp. [1]

Before (including deletions): that converts a scanned schematic into a SPICE netlist through component detection, an

After: for this structural connectivity task. An

Change 6: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: extractor masks components before binarization and returns

Change 7: replace | old pp. [1] | new pp. [1]

Before (including deletions): extractor,

After: segments with explicit endpoints. A typed endpoint graph links endpoints, pins and wire bodies under scale-relative tolerances,

Change 8: replace | old pp. [1] | new pp. [1]

Before (including deletions): graph-based

After: degree-budget completion step reconnects floating pins by bounded assignment. We evaluate on two real- image connectivity benchmarks from CGHD-1152: 31 images with human-verified nets, and 164 held-out photographs from 24 of the corpus's 25 drafters whose reference nets we derive from the dataset's own stroke and symbol annotations. With annotated component boxes, the

Change 9: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: reaches component-pair micro-F1 0.890 on the human-verified set, on which its configuration was selected, and 0.711 on the held-out photographs. On both benchmarks it outperforms every deterministic baseline, including connected-component tracing and Hough line fitting, in paired tests with Holm correction. With annotated wires, its score stays within 0.005 of the native value when the images are rescaled from 0.5× to 3×. Replacing the annotated boxes with a trained component detector gives micro-F1 0.627, and the largest share of

Change 10: replace | old pp. [1] | new pp. [1]

Before (including deletions): resolves electrical connectivity.

After: loss comes from missed components.

Change 11: replace | old pp. [1] | new pp. [1]

Before (including deletions): main contribution

After: wire extractor reaches F1 0.973 on 134 annotated images; on the held-out photographs, wire extraction

Change 12: replace | old pp. [1] | new pp. [1]

Before (including deletions): joining stage: an endpoint-graph

After: main source of error. A vision-language

Change 13: replace | old pp. [1] | new pp. [1]

Before (including deletions): over wire endpoints and component pins with five edge types, combined with degree-budget completion that recovers dropped connections through min-cost b-matching while bounding over-merge with per-pin budgets and self-loop guards. We also introduce a synthetic perturbation protocol and, to our knowledge,

After: given

Change 14: replace | old pp. [1] | new pp. [1]

Before (including deletions): first human-verified net-level connectivity benchmark for hand-drawn circuits, covering 31 images from CGHD-1152. The occlusion-first extractor

After: same electrical-component boxes

Change 15: replace | old pp. [1] | new pp. [1]

Before (including deletions): wire-detection $F1 = 0.976$ on 134 scanned images. On the human-verified net-level benchmark, the best join reaches connectivity micro- $F1 = 0.890$, outperforming the prior completion default (0.83), a legacy radius union-find join (0.67), a connected-component net-tracing baseline (0.62), and a Hough-plus-proximity line tracer (0.81), all under identical detected wires. Running the join on perfect wire labels does not improve micro-F1, showing that connectivity, not wire detection, remains the bottleneck. A first-party comparison shows a strong vision- language model reaches

After: micro-F1

Change 16: replace | old pp. [1] | new pp. [1]

Before (including deletions): same images, statistically indistinguishable

After: human-verified set: the pooled difference

Change 17: replace | old pp. [1] | new pp. [1]

Before (including deletions): but far more expensive

After: is not significant,

Change 18: replace | old pp. [1] | new pp. [1]

Before (including deletions): without structural guarantees, supporting explicit geometric connectivity recovery for deployable circuit digitization.

After: per-image scores favor the model. We release the code, both benchmarks and the evaluation scripts.

Change 19: replace | old pp. [1] | new pp. [1]

Before (including deletions): SPICE,

After: structural netlist,

Change 20: replace | old pp. [1] | new pp. [1]

Before (including deletions): Analog

After: A

Change 21: replace | old pp. [1] | new pp. [1]

Before (including deletions): schematics encode functional information (component

After: schematic encodes component

Change 22: replace | old pp. [1] | new pp. [1]

Before (including deletions): values,

After: component values

Change 23: replace | old pp. [1] | new pp. [1]

Before (including deletions): topological connections)

After: the connections between components

Change 24: replace | old pp. [1] | new pp. [1]

Before (including deletions): visual format designed

After: drawing made

Change 25: replace | old pp. [1] | new pp. [1]

Before (including deletions): human interpretation. Converting scanned

After: people. A netlist records the same connections in a form that software can check, simulate and search. Converting hand-drawn

Change 26: delete | old pp. [1] | new pp. []

Before (including deletions): machine-readable SPICE (Simulation Program with Integrated Circuit Emphasis)

After: (none)

Change 27: replace | old pp. [1] | new pp. [1]

Before (including deletions): enables automated simulation, design verification,

After: would let course material, lab notebooks

Change 28: replace | old pp. [1] | new pp. [1]

Before (including deletions): large-scale circuit analysis. While recent advances

After: legacy designs enter these tools without manual re-entry. This paper addresses the structural part of that conversion: given the components

Change 29: replace | old pp. [1] | new pp. [1]

Before (including deletions): object detection have made component recognition increasingly reliable, the joining problem, determining which wire segments connect to

After: an image, decide

Change 30: replace | old pp. [1] | new pp. [1]

Before (including deletions): pins and to each other, remains the primary failure mode in end-to-end digitization pipelines. The challenge

After: terminals share an electrical node. We call this step joining. Joining

Change 31: replace | old pp. [1] | new pp. [1]

Before (including deletions): threefold.

After: hard for three reasons.

Change 32: delete | old pp. [1] | new pp. []

Before (including deletions): inherently

After: (none)

Change 33: replace | old pp. [1] | new pp. [1]

Before (including deletions): produces endpoint positions

After: yields endpoints at different off-sets from the pins, and strokes stop short of or overshoot their targets. Second, component symbols overlap the wire endpoints

Change 34: replace | old pp. [1] | new pp. [1]

Before (including deletions): vary by 10–80 pixels. Second, components occupy bounding boxes that overlap with wire endpoints, creating ambiguity about whether

After: attach to them, so

Change 35: replace | old pp. [1] | new pp. [1]

Before (including deletions): connects

After: near a component may belong

Change 36: replace | old pp. [1] | new pp. [1]

Before (including deletions): a component's pin

After: one of its pins

Change 37: replace | old pp. [1] | new pp. [1]

Before (including deletions): passes nearby.

After: pass by.

Change 38: replace | old pp. [1] | new pp. [1]

Before (including deletions): fragments,

After: fragments

Change 39: replace | old pp. [1] | new pp. [1]

Before (including deletions): crossing wires create complex connectivity patterns that simple radius-based approaches cannot resolve without over-merging.

After: wire crossings all look alike to a proximity rule.

Change 40: replace | old pp. [2] | new pp. [2]

Before (including deletions): YOLO26m

After: YOLO

Change 41: replace | old pp. [2] | new pp. [2]

Before (including deletions): SPICE

After: Structural

Change 42: replace | old pp. [2] | new pp. [2]

Before (including deletions): Pin discovery + simulation

After: Inferred pin-to-node map

Change 43: replace | old pp. [2] | new pp. [2]

Before (including deletions): End-to-end pipeline

After: Pipeline

Change 44: replace | old pp. [2] | new pp. [2]

Before (including deletions): scanning circuit diagram digitization. Our contributions span the

After: structural circuit-netlist recovery. The highlighted stages extract

Change 45: replace | old pp. [2] | new pp. [2]

Before (including deletions): extraction

After: endpoints

Change 46: delete | old pp. [2] | new pp. []

Before (including deletions): joining stages; the

After: (none)

Change 47: replace | old pp. [2] | new pp. [2]

Before (including deletions): (dashed red) determines which extracted wire segments connect to which component pins and is the dominant failure mode in end-to-end digitization.

After: them into nets.

Change 48: delete | old pp. [2] | new pp. []

Before (including deletions): real

After: (none)

Change 49: replace | old pp. [2] | new pp. [2]

Before (including deletions): circuit images,

After: circuits,

Change 50: replace | old pp. [2] | new pp. [2]

Before (including deletions): our

After: the

Change 51: replace | old pp. [2] | new pp. [2]

Before (including deletions): (connectivity

After: (component-pair

Change 52: replace | old pp. [2] | new pp. [2]

Before (including deletions): (36

After: (35

Change 53: replace | old pp. [2] | new pp. [2]

Before (including deletions): 12 nets); right:

After: 7 nets). Right:

Change 54: replace | old pp. [2] | new pp. [2]

Before (including deletions): (12

After: (11

Change 55: replace | old pp. [2] | new pp. [2]

Before (including deletions): shows, in order: original scan,

After: shows the input,

Change 56: replace | old pp. [2] | new pp. [2]

Before (including deletions): close,

After: closing,

Change 57: replace | old pp. [2] | new pp. [2]

Before (including deletions): (red overlay),

After: (red)

Change 58: replace | old pp. [2] | new pp. [2]

Before (including deletions): every recovered

After: each

Change 59: replace | old pp. [2] | new pp. [2]

Before (including deletions): is a distinct colour drawn along

After: has

Change 60: replace | old pp. [2] | new pp. [2]

Before (including deletions): conductors; component bounding

After: own color;

Change 61: replace | old pp. [2] | new pp. [2]

Before (including deletions): ground-truth labels (as used throughout the net-level evaluation, isolating the join),

After: annotated components,

Change 62: replace | old pp. [2] | new pp. [2]

Before (including deletions): mark junctions,

After: are junctions

Change 63: replace | old pp. [2] | new pp. [2]

Before (including deletions): mark

After: are

Change 64: insert | old pp. [] | new pp. [2]

Before (including deletions): (none)

After: Most image-to-netlist systems erase the components and read nets from the connected components or contours of the remaining wire pixels [1], [2].

Change 65: replace | old pp. [2] | new pp. [2]

Before (including deletions): natural alternative is

After: single stroke gap then splits a net, and a single touching stroke merges two. Other systems pass the image

Change 66: delete | old pp. [2] | new pp. []

Before (including deletions): prompt

After: (none)

Change 67: replace | old pp. [2] | new pp. [2]

Before (including deletions): vision-language

After: language

Change 68: replace | old pp. [2] | new pp. [2]

Before (including deletions): (VLM) to

After: (LLM) and

Change 69: replace | old pp. [2] | new pp. [2]

Before (including deletions): the schematic

After: connectivity from its answer [3]. We keep an explicit representation of every wire endpoint

Change 70: replace | old pp. [2] | new pp. [2]

Before (including deletions): emit its netlist directly. We test this in Section IV-F: a strong VLM attains connectivity accuracy statistically indistinguishable from our

After: decide each connection with

Change 71: replace | old pp. [2] | new pp. [2]

Before (including deletions): pipeline, but it does so at orders-of-magnitude higher cost, returns free-form text rather than a structured netlist, offers no structural guarantee against electrically invalid output (e.g. shorted two-terminal parts), and drops connections on the most complex circuits. These properties motivate an explicit, inspectable geometric pipeline whose output is a netlist by construction and whose accuracy matches the VLM's while being deterministic and structurally valid by construction. We address these challenges with a complete deterministic pipeline whose novel elements are: (1) an

After: rules that can be inspected. This paper makes four contributions: 1) An

Change 72: replace | old pp. [2] | new pp. [2]

Before (including deletions): masks detected components

After: fills each component region with its local median color

Change 73: replace | old pp. [2] | new pp. [2]

Before (including deletions): binarization

After: Sauvola binarization

Change 74: replace | old pp. [2] | new pp. [2]

Before (including deletions): estimates per-segment endpoints, producing a line-segment-with-endpoints representation (rather than tracing nets directly into connected components, as prior systems do) that is what makes graph-based joining possible; (2) an

After: returns line segments whose endpoints are estimated by principal component analysis (PCA). It reaches wire-detection F1 0.973 on 134 CGHD-1152 images (0.976 for the best of 36 variants). 2) A typed

Change 75: delete | old pp. [2] | new pp. []

Before (including deletions): model that constructs a graph over both wire endpoints and component pins,

After: (none)

Change 76: replace | old pp. [2] | new pp. [2]

Before (including deletions): types capturing distinct connectivity patterns;

After: rules, component-first pin assignment

Change 77: replace | old pp. [2] | new pp. [2]

Before (including deletions): (3)

After: scale-relative tolerances, followed by degree-budget completion, which reconnects floating pins through

Change 78: replace | old pp. [2] | new pp. [2]

Before (including deletions): degree-budget completion algorithm that identifies

After: bounded-capacity assignment with a shared-component guard. Disabling two of the edge rules (T-junctions

Change 79: replace | old pp. [2] | new pp. [2]

Before (including deletions): reconnects dropped connections using constrained optimization. We evaluate

After: rail taps), or the directional pin score, has no measurable effect

Change 80: replace | old pp. [2] | new pp. [2]

Before (including deletions): CGHD-1152 dataset [1]

After: 31-image benchmark (Section V-F). 3) Two real-image connectivity benchmarks: 31 CGHD- 1152 images

Change 81: replace | old pp. [2] | new pp. [2]

Before (including deletions): both a synthetic error injection framework that enables systematic robustness analysis

After: human-verified nets,

Change 82: replace | old pp. [2] | new pp. [2]

Before (including deletions): human-verified net-level benchmark.

After: 164- image held-out benchmark covering 24 of the 25 CGHD drafters whose reference nets are derived from the CGHD authors' own stroke-segmentation maps and symbol polygons. The derived reference agrees with the human nets on 17 overlapping images at micro-F1 0.939 and, in an audit of 40 randomly drawn held-out images (37 scored after excluding the three densest; 23 human-checked, 14 settled with a blind model pre- screen), at 0.988 with no false pairs. 4) An evaluation protocol with paired significance tests under Holm correction, complexity and component- size strata, controlled image rescaling, end-to-end runs with a trained component detector, and a vision- language model (VLM) reference that receives the electrical-component boxes given to our join. The output is a structural netlist; recognizing component values and device models is outside the scope of this work. Section II reviews related work, Section III describes the method and Section IV the experimental setup. Section V reports results, and Section VI discusses them together with the scope and limitations of the study.

Change 83: replace | old pp. [2] | new pp. [2]

Before (including deletions): The end-to-end conversion of

After: Several systems convert

Change 84: replace | old pp. [2] | new pp. [2]

Before (including deletions): machine-readable netlists has progressed rapidly.

After: machine- readable netlists.

Change 85: replace | old pp. [2] | new pp. [2]

Before (including deletions): [3]

After: [4]

Change 86: replace | old pp. [2] | new pp. [2]

Before (including deletions): pattern-recognition

After: pattern- recognition

Change 87: replace | old pp. [2] | new pp. [2]

Before (including deletions): digitizing circuit

After: printed

Change 88: replace | old pp. [2] | new pp. [2]

Before (including deletions): com- bining

After: combining

Change 89: replace | old pp. [2] | new pp. [2]

Before (including deletions): [4], the clos- est contemporary system,

After: [1]

Change 90: replace | old pp. [2] | new pp. [2, 3]

Before (including deletions): detector

After: detector, from the You Only Look 2 VOLUME 11, 2023 Once (YOLO) family [5], [6],

Change 91: replace | old pp. [2] | new pp. [3]

Before (including deletions): invokes

After: uses

Change 92: replace | old pp. [2] | new pp. [3]

Before (including deletions): vision-language model (VLM)

After: VLM only

Change 93: replace | old pp. [2] | new pp. [3]

Before (including deletions): reference-designator

After: reference- designator

Change 94: delete | old pp. [2] | new pp. []

Before (including deletions): confines the VLM to symbolic labeling and

After: (none)

Change 95: replace | old pp. [2] | new pp. [3]

Before (including deletions): trusting

After: from

Change 96: replace | old pp. [2] | new pp. [3]

Before (including deletions): VLM with connec- tivity, an insight we share (Section II).

After: VLM.

Change 97: replace | old pp. [2] | new pp. [3]

Before (including deletions): [5] (Auto-SPICE)

After: [3]

Change 98: replace | old pp. [2] | new pp. [3]

Before (including deletions): produce SPICE netlists from

After: generate Simulation Program with Integrated Circuit Empha- sis (SPICE) descriptions of

Change 99: replace | old pp. [2] | new pp. [3]

Before (including deletions): Most directly com- parable to our setting, the

After: The

Change 100: replace | old pp. [2] | new pp. [3]

Before (including deletions): [6] (IEEE Access, 2026)

After: [2]

Change 101: replace | old pp. [2] | new pp. [3]

Before (including deletions): a fully

After: an

Change 102: replace | old pp. [2] | new pp. [3]

Before (including deletions): image-to- SPICE

After: image- to-SPICE

Change 103: replace | old pp. [2] | new pp. [3]

Before (including deletions): circuits, pairing

After: circuits. It pairs

Change 104: replace | old pp. [2] | new pp. [3]

Before (including deletions): module (components

After: module, in which components

Change 105: delete | old pp. [2] | new pp. []

Before (including deletions): 2 VOLUME 11, 2023

After: (none)

Change 106: replace | old pp. [3] | new pp. [3]

Before (including deletions): pixels) and, as we do,

After: pixels, and it

Change 107: delete | old pp. [3] | new pp. []

Before (including deletions): automated

After: (none)

Change 108: replace | old pp. [3] | new pp. [3]

Before (including deletions): simula- tion

After: simulation

Change 109: replace | old pp. [3] | new pp. [3]

Before (including deletions): reference.

After: ref- erence.

Change 110: replace | old pp. [3] | new pp. [3]

Before (including deletions): their

After: an

Change 111: replace | old pp. [3] | new pp. [3]

Before (including deletions): they report

After: it reports

Change 112: replace | old pp. [3] | new pp. [3]

Before (including deletions): whole-netlist

After: whole- netlist

Change 113: replace | old pp. [3] | new pp. [3]

Before (including deletions): Their

After: Its

Change 114: replace | old pp. [3] | new pp. [3]

Before (including deletions): is an instance of

After: belongs to

Change 115: replace | old pp. [3] | new pp. [3]

Before (including deletions): connected-component/contour recipe

After: connected- component family that

Change 116: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: on identical wire inputs

Change 117: replace | old pp. [3] | new pp. [3]

Before (including deletions): IV-D); because their

After: V-C); its

Change 118: delete | old pp. [3] | new pp. []

Before (including deletions): (transistor-level analog, authored in-house)

After: (none)

Change 119: delete | old pp. [3] | new pp. []

Before (including deletions): (LT- spice node-voltage pass/fail)

After: (none)

Change 120: replace | old pp. [3] | new pp. [3]

Before (including deletions): ours (discrete com- ponents on CGHD-1152, component-pair F1),

After: ours, so

Change 121: replace | old pp. [3] | new pp. [3]

Before (including deletions): comparable, but the comparison shows contour/CCL node detection remains the prevailing approach, which our endpoint-graph join substantially outper- forms on a controlled, identical-input benchmark. Instance- segmentation approaches extract circuit graphs directly from pixels:

After: comparable.

Change 122: replace | old pp. [3] | new pp. [3]

Before (including deletions): extraction over

After: ex- traction on

Change 123: replace | old pp. [3] | new pp. [3]

Before (including deletions): in- puts, object-detection

After: inputs, object- detection

Change 124: replace | old pp. [3] | new pp. [3]

Before (including deletions): recent JUHCCR- v1

After: JUHCCR-v1

Change 125: delete | old pp. [3] | new pp. []

Before (including deletions): object

After: (none)

Change 126: delete | old pp. [3] | new pp. []

Before (including deletions): the

After: (none)

Change 127: delete | old pp. [3] | new pp. []

Before (including deletions): family [10], [11]

After: (none)

Change 128: replace | old pp. [3] | new pp. [3]

Before (including deletions): [12], with the ultimate goal of producing simulator-ready descriptions in

After: [10], and target simulator input

Change 129: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: [11]. SPICE-level modeling also supports hardware work in neighboring domains, including memristor-based neuromorphic circuits that realize brain-inspired learning [12] and memristive decision-making circuits for autonomous navigation

Change 130: replace | old pp. [3] | new pp. [3]

Before (including deletions): These works establish detection

After: We focus on the joining stage

Change 131: replace | old pp. [3] | new pp. [3]

Before (including deletions): recognition as largely solved; the bottleneck shifts to reconstructing electrical connectivity, the focus of our contribution.

After: evaluate it with both annotated and detected components.

Change 132: delete | old pp. [3] | new pp. []

Before (including deletions): hand- drawing

After: (none)

Change 133: replace | old pp. [3] | new pp. [3]

Before (including deletions): [1], [2]

After: [14], [15]

Change 134: replace | old pp. [3] | new pp. [3]

Before (including deletions): annotations

After: annotations

Change 135: replace | old pp. [3] | new pp. [3]

Before (including deletions): analog/mixed- signal

After: analog/mixed-signal

Change 136: replace | old pp. [3] | new pp. [3]

Before (including deletions): [14]

After: [16]

Change 137: replace | old pp. [3] | new pp. [3]

Before (including deletions): detec- tion, while

After: detection, and

Change 138: replace | old pp. [3] | new pp. [3]

Before (including deletions): [15] is a benchmark for evaluat- ing

After: [17] evaluates

Change 139: replace | old pp. [3] | new pp. [3]

Before (including deletions): LLM capabilities

After: LLMs

Change 140: delete | old pp. [3] | new pp. []

Before (including deletions): Beyond circuits,

After: (none)

Change 141: replace | old pp. [3] | new pp. [3]

Before (including deletions): [16]

After: [18]

Change 142: replace | old pp. [3] | new pp. [3]

Before (including deletions): an end-to-end

After: a

Change 143: replace | old pp. [3] | new pp. [3]

Before (including deletions): di- agrams,

After: diagrams,

Change 144: replace | old pp. [3] | new pp. [3]

Before (including deletions): an explicit

After: a

Change 145: replace | old pp. [3] | new pp. [3]

Before (including deletions): We use these benchmarks both as evidence of where current systems succeed (component recognition) and where they fail (connectivity), and contribute a synthetic robustness protocol that isolates the join step under controlled detector noise.

After: To our knowledge, no public benchmark reports net-level connectivity on hand- drawn circuit images; Peker et al. [2] report whole-netlist success on an in-house hand-drawn set.

Change 146: replace | old pp. [3] | new pp. [3]

Before (including deletions): [17],

After: [19]

Change 147: replace | old pp. [3] | new pp. [3]

Before (including deletions): understand- ing in particular.

After: understanding.

Change 148: replace | old pp. [3] | new pp. [3]

Before (including deletions): [18]

After: [20]

Change 149: replace | old pp. [3] | new pp. [3]

Before (including deletions): using

After: with

Change 150: delete | old pp. [3] | new pp. []

Before (including deletions): (VQA)

After: (none)

Change 151: replace | old pp. [3] | new pp. [3]

Before (including deletions): connection- matrix

After: connection-matrix

Change 152: delete | old pp. [3] | new pp. []

Before (including deletions): However, multiple independent bench- marks show that general VLMs handle component recogni- tion far better than connectivity.

After: (none)

Change 153: replace | old pp. [3] | new pp. [3]

Before (including deletions): con- nection

After: connection

Change 154: replace | old pp. [3] | new pp. [3]

Before (including deletions): [16], leading

After: [18],

Change 155: delete | old pp. [3] | new pp. []

Before (including deletions): very

After: (none)

Change 156: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 157: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 158: delete | old pp. [3] | new pp. []

Before (including deletions): and

After: (none)

Change 159: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 160: replace | old pp. [3] | new pp. [3]

Before (including deletions): [15]

After: [17]

Change 161: delete | old pp. [3] | new pp. []

Before (including deletions): MLLMs on

After: (none)

Change 162: replace | old pp. [3] | new pp. [3]

Before (including deletions): This corroborates the design choice, shared

After: These results vary

Change 163: replace | old pp. [3] | new pp. [3]

Before (including deletions): SINA [4], to keep wire-to-component

After: task and input, which motivates measuring VLM

Change 164: replace | old pp. [3] | new pp. [3]

Before (including deletions): out of the VLM and instead solve it with an explicit geometric model. We make this separation concrete and quantify it with

After: directly on

Change 165: replace | old pp. [3] | new pp. [3]

Before (including deletions): own Claude-VLM connectivity experiment.

After: benchmark (Section V-I).

Change 166: replace | old pp. [3] | new pp. [3]

Before (including deletions): morphological operations,

After: morphological operations

Change 167: replace | old pp. [3] | new pp. [3]

Before (including deletions): [19], typically preceded by

After: [21], usually after

Change 168: replace | old pp. [3] | new pp. [3]

Before (including deletions): thresholding [20]

After: thresholding [22]

Change 169: replace | old pp. [3] | new pp. [3]

Before (including deletions): [21].

After: [23].

Change 170: replace | old pp. [3] | new pp. [3]

Before (including deletions): dom- inant

After: legacy

Change 171: replace | old pp. [3] | new pp. [3]

Before (including deletions): existing pipelines

After: our pipeline

Change 172: replace | old pp. [3] | new pp. [3]

Before (including deletions): grabs all

After: takes every

Change 173: replace | old pp. [3] | new pp. [3]

Before (including deletions): pins

After: pin

Change 174: replace | old pp. [3] | new pp. [3]

Before (including deletions): (typically 30px), connected via transitive union- find [22]. This over-merges when multiple

After: (30 px), and connections are merged transitively with union-find [24]. When several

Change 175: replace | old pp. [3] | new pp. [3]

Before (including deletions): shorting

After: this rule assigns both pins of a

Change 176: replace | old pp. [3] | new pp. [3]

Before (including deletions): parts (R, C, L, D) onto

After: part to

Change 177: delete | old pp. [3] | new pp. []

Before (including deletions): SINA [4] likewise derives nets from CCL proximity, which is similarly vulnerable to over-merge under detector noise.

After: (none)

Change 178: delete | old pp. [3] | new pp. []

Before (including deletions): instead

After: (none)

Change 179: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: instead

Change 180: replace | old pp. [3] | new pp. [3]

Before (including deletions): incorporates

After: uses

Change 181: replace | old pp. [3] | new pp. [3]

Before (including deletions): geometry

After: geometry,

Change 182: replace | old pp. [3] | new pp. [3]

Before (including deletions): directional preference, and

After: we

Change 183: replace | old pp. [3] | new pp. [3]

Before (including deletions): a degree- budget completion:

After: degree-budget completion,

Change 184: replace | old pp. [3] | new pp. [3]

Before (including deletions): formulation [23], [24]

After: formula- tion [25], [26]

Change 185: replace | old pp. [3] | new pp. [3]

Before (including deletions): per-pin degree and thereby limits over-merge when endpoints are dropped or displaced. This differs from graph

After: the number of recovery connections per pin. Graph

Change 186: replace | old pp. [3] | new pp. [3]

Before (including deletions): engineering [25], which

After: engineer- ing [27]

Change 187: replace | old pp. [3] | new pp. [3]

Before (including deletions): gate-level

After: gate- level

Change 188: replace | old pp. [3] | new pp. [3]

Before (including deletions): a purely geometric model whose

After: geometric, and its

Change 189: replace | old pp. [3] | new pp. [3]

Before (including deletions): relationships

After: relations

Change 190: delete | old pp. [3] | new pp. []

Before (including deletions): wire

After: (none)

Change 191: delete | old pp. [3] | new pp. []

Before (including deletions): com- ponent

After: (none)

Change 192: delete | old pp. [3] | new pp. []

Before (including deletions): A complementary recent direction replaces rule- based connectivity with a learned model:

After: (none)

Change 193: replace | old pp. [3] | new pp. [3]

Before (including deletions): [26] pre- dicts

After: [28] predicts

Change 194: replace | old pp. [3] | new pp. [3]

Before (including deletions): schematics. Such

After: schematics; such learned

Change 195: replace | old pp. [3] | new pp. [3]

Before (including deletions): are powerful but require task-specific connectivity training data and, trained on printed schematics, do not transfer to hand-drawn images without retraining; our geometric model needs no

After: need

Change 196: replace | old pp. [3] | new pp. [3]

Before (including deletions): and is the focus here.

After: data, whereas our join needs none.

Change 197: replace | old pp. [3] | new pp. [3]

Before (including deletions): CircuitNet [27] likewise rely on

After: CircuitNet [29] combine

Change 198: replace | old pp. [3] | new pp. [3]

Before (including deletions): plus

After: with

Change 199: replace | old pp. [3] | new pp. [3]

Before (including deletions): Figure

After: Fig.

Change 200: delete | old pp. [3] | new pp. []

Before (including deletions): six-stage pipeline. The full pipeline proceeds through

After: (none)

Change 201: replace | old pp. [3] | new pp. [3]

Before (including deletions): detection via oriented bounding box

After: detection with an oriented-bounding-box

Change 202: replace | old pp. [3] | new pp. [3]

Before (including deletions): occlusion (filling detected regions

After: occlusion, which fills each component region, enlarged by 15% of its box size (at least 5 px),

Change 203: replace | old pp. [3] | new pp. [3]

Before (including deletions): local

After: the

Change 204: replace | old pp. [3] | new pp. [3]

Before (including deletions): color);

After: color of its surroundings;

Change 205: replace | old pp. [3] | new pp. [3]

Before (including deletions): adaptive binarization;

After: binarization ($k = 0.285$, window 67 px) inside the union of component boxes plus 10 px padding;

Change 206: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: connected-component labeling after a 3×3 elliptical closing, discarding components under 28 px; (5) wire-segment and endpoint estimation by PCA of each

Change 207: replace | old pp. [3] | new pp. [3]

Before (including deletions): component labeling

After: component, followed by overlap deduplication (12■, 8 px) and an anchor filter

Change 208: replace | old pp. [3] | new pp. [3]

Before (including deletions): morphological post-processing; (5) wire end-point extraction via principal component analysis (PCA) on

After: a 16 px

Change 209: replace | old pp. [3] | new pp. [3]

Before (including deletions): regions;

After: distance;

Change 210: delete | old pp. [3] | new pp. []

Before (including deletions): wire

After: (none)

Change 211: replace | old pp. [3] | new pp. [3]

Before (including deletions): via

After: with

Change 212: replace | old pp. [3] | new pp. [3]

Before (including deletions): endpoint-graph model with degree-budget

After: endpoint graph and degree- budget

Change 213: replace | old pp. [3] | new pp. [3]

Before (including deletions): is a netlist mapping

After: maps each

Change 214: replace | old pp. [3] | new pp. [3]

Before (including deletions): pins

After: pin

Change 215: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: an

Change 216: replace | old pp. [3] | new pp. [3]

Before (including deletions): nodes, suitable for SPICE netlist generation. Figure

After: node. Fig.

Change 217: delete | old pp. [3] | new pp. []

Before (including deletions): representative

After: (none)

Change 218: replace | old pp. [3] | new pp. [3]

Before (including deletions): oriented bounding box (OBB) detector

After: OBB detector

Change 219: replace | old pp. [3] | new pp. [3]

Before (including deletions): CGHD-1152 [1], [2].

After: CGHD images [14], [15].

Change 220: replace | old pp. [3] | new pp. [3]

Before (including deletions): encloses

After: encloses

Change 221: delete | old pp. [3] | new pp. []

Before (including deletions): VOLUME 11, 2023 3

After: (none)

Change 222: replace | old pp. [4] | new pp. [3]

Before (including deletions): detector

After: detector

Change 223: replace | old pp. [4] | new pp. [3]

Before (including deletions): CGHD's

After: the

Change 224: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: CGHD

Change 225: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: that

Change 226: replace | old pp. [4] | new pp. [3]

Before (including deletions): distinguish. A

After: distinguish; for example, a

Change 227: replace | old pp. [4] | new pp. [3]

Before (including deletions): light-emitting,

After: light-emitting

Change 228: replace | old pp. [4] | new pp. [3]

Before (including deletions): is a

After: becomes

Change 229: delete | old pp. [4] | new pp. []

Before (including deletions): for the purpose of topology extraction

After: (none)

Change 230: replace | old pp. [4] | new pp. [3]

Before (including deletions): gnd,

After: gnd

Change 231: replace | old pp. [4] | new pp. [3]

Before (including deletions): ab- sorbs

After: absorbs

Change 232: replace | old pp. [4] | new pp. [3]

Before (including deletions): optocoupler,

After: optocoupler VOLUME 11, 2023 3

Change 233: replace | old pp. [4] | new pp. [4]

Before (including deletions): discriminate; these

After: name. The held-out benchmark contains such

Change 234: replace | old pp. [4] | new pp. [4]

Before (including deletions): fall outside

After: as unscored parts, and its extended scoring scores several of them (Section V-D, “Scoring conventions”). Downstream stages use

Change 235: replace | old pp. [4] | new pp. [4]

Before (including deletions): simulatable set

After: class label to construct pins, since the type selects the pin template (for example, capacitor pins lie on the long OBB edges

Change 236: replace | old pp. [4] | new pp. [4]

Before (including deletions): are excluded from

After: resistor pins on

Change 237: replace | old pp. [4] | new pp. [4]

Before (including deletions): net-level bench-

After: short ones), and to

Change 238: replace | old pp. [4] | new pp. [4]

Before (including deletions): in any case. The join model consumes component geometry and pin locations, not device identity, so

After: the inert types that completion skips (Section III-D). Section V-G measures the cost of detector

Change 239: replace | old pp. [4] | new pp. [4]

Before (including deletions): granularity finer than the netlist emits gives the detector further classes to separate while no downstream stage reads the result.

After: errors.

Change 240: replace | old pp. [4] | new pp. [4]

Before (including deletions): diverges

After: differs

Change 241: replace | old pp. [4] | new pp. [4]

Before (including deletions): remaining

After: other

Change 242: replace | old pp. [4] | new pp. [4]

Before (including deletions): leaving

After: and uses

Change 243: replace | old pp. [4] | new pp. [4]

Before (including deletions): random

After: ran- dom

Change 244: replace | old pp. [4] | new pp. [4]

Before (including deletions): images.

After: im- ages; the split manifest was not retained (Section VI-A).

Change 245: replace | old pp. [4] | new pp. [4]

Before (including deletions): 17,

After: size 17 and

Change 246: replace | old pp. [4] | new pp. [4]

Before (including deletions): under

After: with

Change 247: replace | old pp. [4] | new pp. [4]

Before (including deletions): RandAugment,

After: RandAugment

Change 248: replace | old pp. [4] | new pp. [4]

Before (including deletions): HSV jit- ter.

After: hue-saturation-value (HSV) augmentation.

Change 249: replace | old pp. [4] | new pp. [4]

Before (including deletions): held-out validation split

After: val- idation split,

Change 250: replace | old pp. [4] | new pp. [4]

Before (including deletions): check- point attains $\text{mAP}@0.5 = 89.0\%$, $\text{mAP}@0.5:0.95 = 78.5\%$,

After: checkpoint reaches a mean av- erage

Change 251: replace | old pp. [4] | new pp. [4]

Before (including deletions): = 95.8%,

After: (mAP) of 89.0% at an intersection-over- union threshold of 0.5 ($\text{mAP}@0.5$) and 78.5% averaged over thresholds from 0.5 to 0.95 ($\text{mAP}@0.5:0.95$), with precision 95.8%

Change 252: delete | old pp. [4] | new pp. []

Before (including deletions): =

After: (none)

Change 253: replace | old pp. [4] | new pp. [4]

Before (including deletions): which trails

After: 12.4 points below

Change 254: replace | old pp. [4] | new pp. [4]

Before (including deletions): 83.1%) by 12.4 points:

After: 83.1%):

Change 255: replace | old pp. [4] | new pp. [4]

Before (including deletions): are

After: look

Change 256: replace | old pp. [4] | new pp. [4]

Before (including deletions): indistinguishable from an ordinary junction-free

After: like any other wire

Change 257: delete | old pp. [4] | new pp. []

Before (including deletions): 61-to-16

After: (none)

Change 258: delete | old pp. [4] | new pp. []

Before (including deletions): a set of

After: (none)

Change 259: replace | old pp. [4] | new pp. [4]

Before (including deletions): where

After: with

Change 260: replace | old pp. [4] | new pp. [4]

Before (including deletions): $\cup P$.

After: $\cup P$ and five edge types.

Change 261: replace | old pp. [4] | new pp. [4]

Before (including deletions): with weight 0. This ensures

After: so that

Change 262: replace | old pp. [4] | new pp. [4]

Before (including deletions): of a wire are always in

After: belong to

Change 263: replace | old pp. [4] | new pp. [4]

Before (including deletions): endpoint pairs (ei, ej)

After: endpoints ei, ej on different wires

Change 264: replace | old pp. [4] | new pp. [4]

Before (including deletions): edge with weight $\blacksquare_{ei} - ej \blacksquare$.

After: an edge.

Change 265: replace | old pp. [4] | new pp. [4]

Before (including deletions): captures wire

After: rule connects

Change 266: replace | old pp. [4] | new pp. [4]

Before (including deletions): corners,

After: corners

Change 267: replace | old pp. [4] | new pp. [4]

Before (including deletions): For each

After: Each

Change 268: replace | old pp. [4] | new pp. [4]

Before (including deletions): e

After: connects to at most one pin. The component-first assignment described below chooses the pin. If no component is assigned, the endpoint takes the pin within $d \leq \tau_{\text{pin}}$ with the smallest score, where the score is $d[1 - \alpha \max(0, \cos \theta)]$ with $\alpha = 0.35$

Change 269: delete | old pp. [4] | new pp. []

Before (including deletions): pin p with $\|e - p\| \leq \tau_{\text{pin}}$, add edge with weight $\|e - p\| \cdot (1 - \alpha \cdot \cos \theta)$, where

After: (none)

Change 270: delete | old pp. [4] | new pp. []

Before (including deletions): is

After: (none)

Change 271: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: outward

Change 272: replace | old pp. [4] | new pp. [4]

Before (including deletions): vector, and α controls directional preference. This

After: vector (for $d \leq 10 - 6 \text{ px}$ the score

Change 273: replace | old pp. [4] | new pp. [4]

Before (including deletions): the critical

After: d). If neither step finds a pin, no

Change 274: replace | old pp. [4] | new pp. [4]

Before (including deletions): type: it connects wires to components.

After: is added.

Change 275: replace | old pp. [4] | new pp. [4]

Before (including deletions): each

After: an

Change 276: replace | old pp. [4] | new pp. [4]

Before (including deletions): wire

After: a different

Change 277: replace | old pp. [4] | new pp. [4]

Before (including deletions): $w_j = (c, d)$,

After: w_j , add an edge

Change 278: replace | old pp. [4] | new pp. [4]

Before (including deletions): $\text{dist}(e, w_j) \leq \tau$, add edge with weight $\text{dist}(e, w_j)$.

After: is at most τ and the normalized projection parameter lies in $[0.15, 0.85]$.

Change 279: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: rule

Change 280: replace | old pp. [4] | new pp. [4]

Before (including deletions): where a wire terminates on another wire's body.

After: away from segment ends.

Change 281: replace | old pp. [4] | new pp. [4]

Before (including deletions): each component

After: a

Change 282: replace | old pp. [4] | new pp. [4]

Before (including deletions): wire

After: a

Change 283: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: add an edge

Change 284: replace | old pp. [4] | new pp. [4]

Before (including deletions): $\text{dist}(p, w_j) \leq \tau$

After: is at most τ

Change 285: replace | old pp. [4] | new pp. [4]

Before (including deletions): lands mid-span (not at w_j 's endpoints, which are handled by edge type 3), add edge with weight $\text{dist}(p, w_j)$.

After: parameter lies in $[0.05, 0.95]$.

Change 286: replace | old pp. [4] | new pp. [4]

Before (including deletions): binds components that tap

After: rule captures pins tapped

Change 287: replace | old pp. [4] | new pp. [4]

Before (including deletions): passing rail along

After: rail. (a) Radius-based joining R1 C1 R2 radius grabs every pin in range $\rightarrow$ R1, C1, R2 wrongly merged (b)

Endpoint-graph (ours) R1 C1 R2 Wire body Endpoint-pin Endpoint-endpoint T-junction (endpt-body) Rail-tap (pin-body)

Endpoint Pin FIGURE 3. Joining approaches. (a) Radius-based join (legacy): each wire endpoint takes all pins within a fixed threshold, which over-merges. (b) Endpoint-graph model: typed edges and component-first pin assignment; directional scoring

applies only to

Change 288: replace | old pp. [4] | new pp. [4]

Before (including deletions): wire's body rather than at its end. It recovers connectivity that endpoint-only binding misses, without the all-to-all over-merge of radius grabbing.

After: fallback pin search.

Change 289: replace | old pp. [4] | new pp. [4]

Before (including deletions): τ_{join} , τ_{pin} ,

After: are scale-relative with fixed pixel clamps: $\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$, $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$

Change 290: delete | old pp. [4] | new pp. []

Before (including deletions): are scale-relative: τ

After: (none)

Change 291: replace | old pp. [4] | new pp. [4]

Before (including deletions): $k \cdot s$ where

After: $\min(20, \max(8, 0.20s))$, in pixels. Here

Change 292: replace | old pp. [4] | new pp. [4]

Before (including deletions): characteristic

After: median component-box diagonal of the image (the upper middle value for an even count), with a wire-length

Change 293: replace | old pp. [4] | new pp. [4]

Before (including deletions): (median

After: as fallback when

Change 294: replace | old pp. [4] | new pp. [4]

Before (including deletions): diagonal), so

After: geometry is unavailable. The multipliers and clamps are set by hand. Because s is one scalar per image, these three tolerances do not vary with component size inside

Change 295: replace | old pp. [4] | new pp. [4]

Before (including deletions): single parameter setting works across circuits of varying size.

After: drawing.

Change 296: delete | old pp. [4] | new pp. []

Before (including deletions): we use a two-step assignment. First, assign

After: (none)

Change 297: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: is first assigned

Change 298: replace | old pp. [4] | new pp. [4]

Before (including deletions): its

After: a component. Its distance to a component is 0 inside the component's OBB and otherwise the distance to the nearest OBB edge (axis-aligned box distance when OBB vertices are unavailable). The endpoint is assigned to the

Change 299: replace | old pp. [4] | new pp. [4]

Before (including deletions): by bounding-box distance (0 if inside the bbox, using a generous radius of

After: within

Change 300: replace | old pp. [4] | new pp. [4]

Before (including deletions): $\cdot$ diagonal)). Second, within

After: δc), where δc is that component's own box diagonal. Because this radius grows with each component's diagonal, endpoint- to-component binding follows the size of each component in addition to the image-level scale s . Within

Change 301: delete | old pp. [4] | new pp. []

Before (including deletions): select

After: (none)

Change 302: delete | old pp. [4] | new pp. []

Before (including deletions): pin by geometric position: for horizontal components ($w > h$), left pin if

After: (none)

Change 303: replace | old pp. [4] | new pp. [4]

Before (including deletions): $x < \text{center } x$, else right pin; for vertical components ($h > w$), top pin if endpoint $y < \text{center } y$, else bottom

After: takes the nearest

Change 304: replace | old pp. [4] | new pp. [4]

Before (including deletions): This handles the common case where endpoints land inside a component's bounding box but far from its pin. Figure

After: Fig.

Change 305: replace | old pp. [4] | new pp. [4]

Before (including deletions): our endpoint- graph model.

After: the endpoint graph.

Change 306: replace | old pp. [4] | new pp. [4]

Before (including deletions): computed as

After: the

Change 307: replace | old pp. [4] | new pp. [4]

Before (including deletions): projected

After: accumulated with union-find and projected

Change 308: replace | old pp. [4] | new pp. [4]

Before (including deletions): is a netlist mapping

After: maps

Change 309: replace | old pp. [4] | new pp. [4]

Before (including deletions): The endpoint-graph join may leave

After: After base joining,

Change 310: delete | old pp. [4] | new pp. []

Before (including deletions): component

After: (none)

Change 311: replace | old pp. [4] | new pp. [4]

Before (including deletions): “floating”: touching

After: lie in a net that contains

Change 312: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: pins of

Change 313: delete | old pp. [4] | new pp. []

Before (including deletions): component's net, with no wire connecting them to another

After: (none)

Change 314: replace | old pp. [4] | new pp. [4]

Before (including deletions): This indicates a dropped or over-displaced wire endpoint that the detector failed to place near the correct pin. Degree-budget completion identifies

After: Completion treats

Change 315: delete | old pp. [4] | new pp. []

Before (including deletions): and reconnects them via constrained optimization. A pin p is treated

After: (none)

Change 316: replace | old pp. [4] | new pp. [4]

Before (including deletions): floating if its net contains only pins from p's own component,

After: candidates for recovery,

Change 317: replace | old pp. [4] | new pp. [4]

Before (including deletions): such as gnd, vss,

After: (ground, supply markers,

Change 318: replace | old pp. [4] | new pp. [4]

Before (including deletions): antenna, junction,

After: antennas, probes, junctions

Change 319: replace | old pp. [4] | new pp. [4]

Before (including deletions): terminal. For

After: terminals). Completion keeps the base nets and proposes connections from

Change 320: replace | old pp. [4] | new pp. [4]

Before (including deletions): p, we generate candidate edges

After: p

Change 321: delete | old pp. [4] | new pp. []

Before (including deletions): all

After: (none)

Change 322: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: of other components

Change 323: replace | old pp. [4] | new pp. [4]

Before (including deletions): components within

After: nets. A candidate's distance must not exceed the

Change 324: replace | old pp. [4] | new pp. [4]

Before (including deletions): $p \cdot s$,

After: $p \tau c$,

Change 325: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: $\tau c = \min(60, \max(24, 0.62s)) p x$ and

Change 326: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: = 4. For each candidate we look for a wire witness: a detected segment whose endpoints lie near p and q in either orientation. The witness cost

Change 327: replace | old pp. [4] | new pp. [4]

Before (including deletions): reach factor; we use $p = 4$ for

After: smallest, over segments and orientations, of

Change 328: replace | old pp. [4] | new pp. [4]

Before (including deletions): deployed scale-relative join, and Section IV-D shows

After: larger endpoint-to-pin distance, bounded by r . Without

Change 329: replace | old pp. [4] | new pp. [4]

Before (including deletions): broad F1 plateau for $p \in [3, 5]$. Candidate edge weight

After: witness, the candidate cost

Change 330: replace | old pp. [4] | new pp. [4]

Before (including deletions): Euclidean distance $\|p - q\|$. We then solve

After: $\|p - q\|$. The candidates form

Change 331: replace | old pp. [4] | new pp. [4]

Before (including deletions): min-cost b-matching problem, $\min P(p, q) \in E' w(p, q) \cdot x_{pq}$ subject to $P_{pq} x_{pq} \leq 1$ for each

After: rectangular assignment matrix with one row per

Change 332: replace | old pp. [4] | new pp. [4]

Before (including deletions): pin p , using the Hungarian algorithm [23] (`scipy.optimize.linear_sum_assignment`).

After: pin.

Change 333: replace | old pp. [4] | new pp. [4]

Before (including deletions): offers

After: supplies

Change 334: replace | old pp. [4] | new pp. [4]

Before (including deletions): slots so legitimate multi-terminal nets are not starved, and a per-pin opt-out leaves a pin with no good target unconnected rather than forcing a spurious edge. Finally, we reject any matched edge (p, q) if p and q belong to the same component or if adding the edge would merge two nets that already share a component, which structurally prevents short-circuiting two-terminal parts. Completion edges are “wireless”: they represent inferred connections without corresponding wire segments, analogous to manual pin merges in schematic editors. Figure 4 illustrates the completion step.

After: slot columns, limited by its can-

Change 335: replace | old pp. [5] | new pp. [5]

Before (including deletions): Merges: The

After: merges from the

Change 336: replace | old pp. [5] | new pp. [5]

Before (including deletions): Component Classes Reduced

After: component classes

Change 337: replace | old pp. [5] | new pp. [5]

Before (including deletions): Eight Merged Targets Are Listed With Their Source Classes in Full; a Further Seven Classes Pass Through Unchanged (terminal, crossover, switch, text, junction, gnd, vss)

After: eight merged targets are listed with their source classes; seven further classes pass through unchanged

Change 338: replace | old pp. [5] | new pp. [5]

Before (including deletions): Remaining

After: remaining

Change 339: replace | old pp. [5] | new pp. [5]

Before (including deletions): Fold Into

After: fold into

Change 340: delete | old pp. [5] | new pp. []

Before (including deletions): Total: 24 source classes merged into 8 targets. capacitor, transistor, and voltage_source have no bare counterpart in CGHD: they exist only as subtypes, and voltage_source renames the voltage.* family. (a) Radius-based joining R1 C1 R2 radius grabs every pin in range →R1, C1, R2 wrongly merged (b) Endpoint-graph (ours) R1 C1 R2 Wire body Endpoint–pin Endpoint–endpoint T-junction (endpt–body) Rail-tap (pin–body) Endpoint Pin FIGURE 3. Joining approaches compared. (a) Radius-based join (legacy): each wire endpoint grabs all pins within a fixed threshold, causing over-merging. (b) Our endpoint-graph model: typed edges (wire body, endpoint–pin) with directional preference resolve connectivity without over-merging.

After: (none)

Change 341: delete | old pp. [5] | new pp. []

Before (including deletions): self-loop

After: (none)

Change 342: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: no merge between nets that already share

Change 343: replace | old pp. [5] | new pp. [5]

Before (including deletions): component's own two terminals are never matched together

After: component

Change 344: replace | old pp. [5] | new pp. [5]

Before (including deletions): After

After: after

Change 345: replace | old pp. [5] | new pp. [5]

Before (including deletions): “floating” (touching

After: floating (connected

Change 346: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: to

Change 347: replace | old pp. [5] | new pp. [5]

Before (including deletions): Completion

After: completion

Change 348: replace | old pp. [5] | new pp. [5]

Before (including deletions): via

After: by

Change 349: replace | old pp. [5] | new pp. [5]

Before (including deletions): dashed),

After: dashed); a guard rejects merges between nets that already share a component. didate demand, and each row has a private dummy column of cost 1.5r for opting out. Unavailable as- signments cost 106. The Hungarian algorithm [25] (`scipy.optimize.linear_sum_assignment`) as- signs every row to one column,

Change 350: replace | old pp. [5] | new pp. [5]

Before (including deletions): self-loop

After: at most one row per column; the target slots implement bounded-capacity match- ing [26]. Matches are applied in sequence. Dummy and unavailable assignments add no edge. A merge is skipped if both pins are already in the same net or if their nets already share a component; this

Change 351: replace | old pp. [5] | new pp. [5]

Before (including deletions): preventing shorts

After: acts on completion merges only. Com- pletion edges have no corresponding wire segment, much like a manual pin merge in a schematic editor. Fig. 4 illustrates the step. IV. EXPERIMENTAL SETUP A. DATASETS AND IMAGE PROVENANCE All real images come from CGHD-1152 [14], [15], a public corpus of hand-drawn circuit images organized by drafter. We use four evaluation sets. The wire benchmark has 134 images with human wire labels and annotated component boxes. The human-verified net benchmark has 31 images from 10 drafters. The 34 annotated images were chosen by circuit content, not appearance: ten initial images spanning a range of complexity, and the rest sampled at even intervals of electrical-component count from the circuits whose parts all belong to the scored device types. Image background was not a selection criterion, but the selected copies have mostly plain backgrounds, partly because 16 of them are stroke maps (see below). Three annotated images were excluded before scoring (one unverified, two failing physical-sanity checks). Net proposals were bootstrapped by a conservative endpoint- graph join over the annotated wires and corrected by one annotator in a purpose-built verification tool. The 31 images contain 3–14 electrical components each (median 7); 15 have at most five and 12 have at least ten. The independent held-out benchmark (164 images, 24 drafters) and the synthetic suite (15 authored circuits) are described in Sections V-D and V-A. The 134 wire-benchmark and 31 net-benchmark images are 704 × 704 copies from a Roboflow export of CGHD. The export resized each original without preserving its aspect ratio. Of the 31 net-benchmark images, 14 are also flipped or rotated relative to the CGHD file, and 16 (45 of the 134 wire- benchmark images) are CGHD’s binary stroke-segmentation renderings instead of photographs. Wire and net labels were drawn on these copies. The held-out benchmark uses the original CGHD photographs. B. CONNECTIVITY METRIC For each image we form the set of unordered pairs of electrical components that share at least one net, and compare predicted with reference pairs. The electrical subset contains resistors, capacitors, inductors, diodes, transistors, sources and inte- grated circuits (ICs); junctions,

terminals, ground symbols and crossovers affect the score only through the connections they mediate. The metric ignores which terminal of a component carries a connection. Micro-F1 pools true positives (TP), false positives (FP) and false negatives (FN) over images, so a dense board contributes in proportion to its connections; it is our primary metric. Macro-F1, the mean of per-image F1, is reported alongside. C. STATISTICAL PROTOCOL All comparisons are paired over images. We report 95% percentile confidence intervals (CIs) from an image-level bootstrap with $B = 10,000$ resamples, and test each difference with three procedures: the bootstrap itself, a sign-flip permutation test that swaps the two methods' outputs

Change 352: replace | old pp. [5] | new pp. [5]

Before (including deletions): two-terminal components. IV. RESULTS A. SYNTHETIC EVALUATION FRAMEWORK We author 15 synthetic circuits spanning 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops,

After: each image (exact when at most 22 images differ, otherwise 100,000 Monte Carlo draws),

Change 353: replace | old pp. [5] | new pp. [5]

Before (including deletions): 6- component ring (cyclic and bridge topologies being

After: Wilcoxon signed-rank test on per-image F1. The first two test

Change 354: replace | old pp. [5] | new pp. [5]

Before (including deletions): main stress cases

After: micro-F1 difference; the Wilcoxon test compares images. Holm's correction is applied within each procedure across the comparisons of a table. For the human-verified benchmark we report all three Holm-adjusted p values (Table 5);

Change 355: replace | old pp. [5] | new pp. [5]

Before (including deletions): join robustness). Each circuit has a known ground-truth netlist and SPICE deck with a test voltage source. We inject five categories of error at four severity levels (L1–L4) on top

After: the held-out benchmark we report the larger

Change 356: replace | old pp. [5] | new pp. [5]

Before (including deletions): clean (L0) baseline. These are Gaussian endpoint jitter with standard deviation σ (from 3 px at L1 to 14 px at L4), cut-short corruption in which each wire is shortened inward by a fixed length to model endpoints that stop short of the pin (from 4 px at L1 to 22 px at L4), full wire drops with probability p_{drop} (up to 0.20 at L4), wrong- pin snaps with probability p_{wp} (from 0.02 at L2 to 0.10 at L4),

After: Holm-adjusted bootstrap

Change 357: replace | old pp. [5] | new pp. [5]

Before (including deletions): anchor displacement, where with probability p_{disp} one endpoint is moved by up to a_{max} pixels (from 30 px at L1 to 80 px at L4). We report join F1 as the component-connectivity F1 score, measuring which pairs of components share a net in the predicted versus ground-truth netlist. We report simulation accuracy as the percentage of random seeds for which every source current in the predicted SPICE netlist matches the clean oracle within 5%. B. WIRE DETECTION BENCHMARK We evaluate on 134 images from CGHD-1152 with ground- truth wire and component labels. Figure 5 and Table 2 summarize the top configurations across 36 tested variants. TABLE 2. Wire Detection Performance (134 images) Configuration F1 Prec Rec Sauvola + anchor 16 px (ours) 0.976 0.973 0.978 Sauvola + anchor 12 px 0.973 0.974 0.972 Sauvola (w=61) + anchor 12 px 0.959 0.944 0.974 Sauvola (k=0.2875) + anchor 14 px 0.950 0.921 0.980 Otsu threshold + CCL 0.789 0.796 0.783

After: per-

Change 358: replace | old pp. [6] | new pp. [6]

Before (including deletions): 0.6 0.7 0.8 0.9 1.0 Wire detection

After: TABLE 2. Synthetic join leaderboard: mean component-pair

Change 359: replace | old pp. [6] | new pp. [6]

Before (including deletions): (134 images) Triangle Otsu adaptive Gaussian Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola, anchor 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.845 0.949 0.950 0.959 0.973 0.976 FIGURE 5. Wire

detection F1 across thresholding methods and pipeline variants (134 images). Our configuration (Sauvola binarization with a 16 px anchor filter) achieves $F1 = 0.976$. Otsu and triangle thresholding degrade significantly. The pipeline achieves median $F1 = 1.000$ across the benchmark set, with 87% of images scoring $F1 \geq 0.90$. Sauvola adaptive binarization dominates all alternatives: Otsu yields $F1 = 0.789$, adaptive Gaussian $F1 = 0.845$. Component occlusion matters: without it, component internals (text, symbols, internal wiring) generate hundreds of false wire detections. C. JOIN STRATEGY COMPARISON Table 3 and Figure 6 compare joining strategies on the synthetic evaluation framework (15

After: over 15

Change 360: replace | old pp. [6] | new pp. [6]

Before (including deletions): seeds, mean F1 over

After: seeds at each

Change 361: replace | old pp. [6] | new pp. [6]

Before (including deletions): levels L1–L4). TABLE 3. Join Strategy Leaderboard (Synthetic Eval)

After: level.

Change 362: replace | old pp. [6] | new pp. [6]

Before (including deletions): The completion-based strategies are

After: mutation p values (Table 6), and

Change 363: replace | old pp. [6] | new pp. [6]

Before (including deletions): most robust, maintaining $F1 \geq 0.94$ at the highest error severity (L4), compared to $F1 = 0.36$

After: Wilcoxon test is reported

Change 364: replace | old pp. [6] | new pp. [6]

Before (including deletions): legacy radius union-find baseline, a $2.6\times$ improvement. The radius join's transitive grabbing causes catastrophic over-merging under noise: at L4, it shorts multiple components onto shared nets.

After: human-verified benchmark only.

Change 365: replace | old pp. [6] | new pp. [6]

Before (including deletions): emphasize, however, that the endpoint-graph base joins are themselves far more robust than the radius join, reaching $F1 = 0.90$ at L4 (Table 3). The degree-budget completion adds a further $+0.05$ over the strongest of these by recovering dropped connections the base graph

After: also count per-image wins, ties and losses (W/T/L). Unless stated otherwise (Section V-G), every

Change 366: replace | old pp. [6] | new pp. [6]

Before (including deletions): leaves floating, while its per-pin edge budget keeps precision high. Pairing that completion with the high-precision scale-relative base (the scale-relative completion join used throughout) gives the best synthetic robustness (0.95 at L4) and, more importantly, the best real-image connectivity (Section IV-D). The large gap is against the legacy radius join; against the strongest prior graph join the improvement is smaller but consistent across every severity level. For readability we omit two intermediate variants whose synthetic scores fall within rounding of rows shown (nearest-pin without anchoring,

After: receives annotated component boxes

Change 367: delete | old pp. [6] | new pp. []

Before (including deletions): the scale-relative base with end-extension but no dead-end rescue). D. REAL-IMAGE NET-LEVEL EVALUATION The synthetic results above measure robustness under a parametric error model. To validate on real detector output, we constructed, to our knowledge, the first human-verified net-level ground truth for hand-drawn circuits: 31 images from CGHD-1152 in which every electrical net was traced and confirmed by a human annotator through a purpose-built verification tool. Because one of our comparison points is a vision-language model (Section IV-F), the human, not any model, is the verifier of record, so the answer key is independent of every method under test. Net-level proposals were bootstrapped by a conservative endpoint-graph join over the dataset's perfect wire labels and then corrected by hand; during verification the human consistently caught cases where the bootstrap join had wrongly split a single rail, and we additionally

re-validate all conclusions on the independently authored synthetic ground truth to rule out residual bootstrap bias. Connectivity is scored as component-pair F1 restricted to SPICE-active components, identical to the synthetic metric. Table 4 reports every registered join strategy on the verified set, run on

After: (none)

Change 368: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: V. RESULTS A. SYNTHETIC EVALUATION

Change 369: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: authored 15 circuits with 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops and a 6-component ring. Each circuit has a known netlist, component values and a test voltage source. We inject five error types at four severity levels (L1–L4) on top of the clean level L0: Gaussian endpoint jitter (σ from 3 px at L1 to 14 px at L4); cut-short wires, shortened inward by 4 px at L1 to 22 px at L4; wire drops with probability up to 0.20 at L4; wrong-pin snaps with probability from 0.02 at L2 to 0.10 at L4; and anchor displacement, which moves one endpoint by up to 30 px at L1 and 80 px at L4. Besides component-pair F1, we

Change 370: replace | old pp. [6] | new pp. [6]

Before (including deletions): micro-averaged (pair-pooled) component-pair

After: simulation accuracy: the percentage of seeds for which every source current of the predicted netlist, simulated with the authored values, matches the clean reference within 5%. Table 2 and Fig. 5 compare the join strategies. At L4, the scale-relative graph with completion scores

Change 371: replace | old pp. [6] | new pp. [6]

Before (including deletions): as

After: 0.95,

Change 372: replace | old pp. [6] | new pp. [6]

Before (including deletions): primary metric,

After: rescue graph with completion 0.94, the rescue graph alone 0.90, the scale-relative graph alone 0.85 and radius union-find 0.36. The rescue graph differs from the scale-relative graph by a 12 px end-extension and a dead-end rescue step that gives dangling wire ends a longer directional reach. Table 3 breaks down our join by circuit at L4. Precision is at least 0.85 on every circuit. The Wheatstone bridge is the hardest case, with F1 0.82 and simulation accuracy 6%; the parallel R–R circuit keeps F1 1.00 at 25% simulation accuracy,

Change 373: replace | old pp. [6] | new pp. [6]

Before (including deletions): that

After: the two measures capture different properties. B. WIRE DETECTION Table 4 and Fig. 6 summarize the best of 36 extractor variants on the 134-image wire benchmark. Sauvola binarization with

Change 374: replace | old pp. [6] | new pp. [6]

Before (including deletions): dense 14-component board contributes in proportion to its connections rather than counting

After: 16 px anchor filter reaches F1 0.976 (median per-image F1 1.000; 87% of images at or above 0.90). Otsu thresholding reaches 0.789 and adaptive Gaussian thresholding 0.845. This best variant deduplicates overlapping segments at 10/18 px; the deployed setting, which uses 12/8 px with

Change 375: replace | old pp. [6] | new pp. [6]

Before (including deletions): as a 3-component one;

After: anchor filter, scores 0.973 and is

Change 376: replace | old pp. [6] | new pp. [6]

Before (including deletions): macro (mean per-image)

After: extractor used in every other TABLE 3. Scale-relative completion join on eight of the 15 synthetic circuits at L4 (16 seeds). Circuit Comps F1 Prec Sim. acc. Parallel R–R 3 1.00 1.00 25% R–R divider 3 0.99 1.00 69% RL series 3 0.99 1.00 81% Diode–R loop 3 0.99 1.00 69% Dual R loops 4 0.93 0.90 12% 4-component loop 4 0.94 0.91 56% Wheatstone bridge 6 0.82 0.85 6% 6-component ring 6 0.95 0.92 50% TABLE 4. Wire detection on 134 images with annotated component boxes. The deployed row uses 12█/8 px overlap deduplication and is the extractor used in all other experiments; the other Sauvola rows use 10█/18 px. Configuration F1 Prec Rec Sauvola + anchor 16 px, 12█/8 px dedup (deployed) 0.973 0.967 0.978 Sauvola + anchor 16 px (best variant) 0.976 0.973 0.978 Sauvola + anchor 12 px 0.973 0.974 0.972 Sauvola (w=61) + anchor 12 px 0.959 0.944 0.974 Sauvola (k=0.2875) + anchor 14 px 0.950 0.921 0.980 Otsu threshold + CCL 0.789 0.796 0.783 experiment. The score does not depend on image modality:

Change 377: replace | old pp. [6] | new pp. [6]

Before (including deletions): listed alongside for reference. The best configuration,

After: 0.967 on the 89 photographs and 0.986 on the 45 stroke maps. C. HUMAN-VERIFIED BENCHMARK Table 5 reports every join configuration and baseline on the 31-image human-verified benchmark, together with the paired tests, and Fig. 7 plots the micro-F1 values with their CIs. Our join, degree-budget completion on

Change 378: replace | old pp. [6] | new pp. [6]

Before (including deletions): completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves

After: base, reaches

Change 379: delete | old pp. [6] | new pp. []

Before (including deletions): =

After: (none)

Change 380: replace | old pp. [6] | new pp. [6]

Before (including deletions): = 0.92,

After: 0.919,

Change 381: replace | old pp. [6] | new pp. [6]

Before (including deletions): = 0.86;

After: 0.864; macro- F1 0.901; TP/FP/FN 418/37/66;

Change 382: delete | old pp. [6] | new pp. []

Before (including deletions): bootstrap

After: (none)

Change 383: replace | old pp. [6] | new pp. [6]

Before (including deletions): 0.924]), improving

After: 0.924]). It beats every deterministic comparator

Change 384: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: every test after Holm correction. The margin over

Change 385: replace | old pp. [6] | new pp. [6]

Before (including deletions): default that

After: default, which

Change 386: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: base, is +0.061 [+0.025, +0.100] with 11 wins, 18 ties and 2 losses. Completion lifts the scale-relative

Change 387: replace | old pp. [6] | new pp. [6]

Before (including deletions): (0.829) and the radius union-find join (0.667) (Figure 7).

After: from 0.816 to 0.890, mostly through recall (0.727 to 0.864).

Change 388: replace | old pp. [6] | new pp. [6]

Before (including deletions): gain over that rescue-based completion comes from the cleaner base: the rescue base's 12 px end-extension and dead-end rescue over-reach on noisy detected wires and cost precision, whereas the scale-relative

After: scale-relative

Change 389: delete | old pp. [6] | new pp. []

Before (including deletions): keeps precision high while the bounded completion still recovers recall. A reach sweep shows a broad F1 plateau across reach 3–5×s, so the setting

After: (none)

Change 390: replace | old pp. [6] | new pp. [6]

Before (including deletions): robust rather

After: also more precise

Change 391: delete | old pp. [6] | new pp. []

Before (including deletions): tuned to a spike. Confidence intervals are wide at $N = 31$ and the per-strategy intervals overlap, but the ranking is consistent and is corroborated on the independently authored synthetic suite (Table 3). This real-image ranking resolves an open question from prior work on this pipeline: against bootstrapped pilot labels the degree-budget completion had not consistently beaten its base join; against human-verified ground truth it clearly does (micro-F1 0.829 vs 0.787 for

After: (none)

Change 392: replace | old pp. [6] | new pp. [6]

Before (including deletions): alone),

After: (0.929 versus 0.811),

Change 393: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: completion keeps most of that precision (0.919, against 0.789 for completion on

Change 394: replace | old pp. [6] | new pp. [6]

Before (including deletions): scale-relative variant is better still. The same ordering holds

After: rescue base). Leaving out any one image moves our micro-F1 within 0.885–0.900, and no deletion changes the sign of any comparison. A sweep of the completion reach over $p \in [3, 5]$ gives macro-F1 0.895–0.903. Several image-to-netlist systems erase component regions and read nets from the remaining wire pixels with connected components or contours [1], [2]. On our detected wires, the best connected-component configuration reaches micro-F1 0.624 at precision 0.965: it rarely merges wrongly, but every stroke gap splits a net (recall 0.461). Hough line fitting [21] with proximity linking reaches 0.805 at its best linking tolerance. Both baselines use the best configuration of their sweep

Change 395: replace | old pp. [7] | new pp. [7]

Before (including deletions): 6.

After: 5.

Change 396: replace | old pp. [7] | new pp. [7]

Before (including deletions): strategy robustness

After: strategies

Change 397: delete | old pp. [7] | new pp. []

Before (including deletions): bars give

After: (none)

Change 398: replace | old pp. [7] | new pp. [7]

Before (including deletions): connectivity

After: component-pair

Change 399: delete | old pp. [7] | new pp. []

Before (including deletions): The completion-based strategies (left) stay near 1.0, while the radius union-find and nearest-pin baselines (right) collapse at L3–L4.

After: (none)

Change 400: replace | old pp. [7] | new pp. [7]

Before (including deletions): connectivity

After: component-pair

Change 401: replace | old pp. [7] | new pp. [7]

Before (including deletions): Errors cause fragmentation rather than shorts (precision stays high; Table 5);

After: TABLE 5. Component-pair connectivity on

Change 402: replace | old pp. [7] | new pp. [7]

Before (including deletions): balanced Wheatstone bridge is

After: 31-image human-verified benchmark (annotated component boxes; detected wires unless noted). F1, P and R are micro-averaged. Δ is

Change 403: replace | old pp. [7] | new pp. [7]

Before (including deletions): hardest case. TABLE 4. Join Strategy

After: micro-F1 of ours minus the comparator with 95% paired bootstrap CI; Holm-adjusted p values are given for the bootstrap, the permutation test on micro-F1

Change 404: replace | old pp. [7] | new pp. [7]

Before (including deletions): Baseline Comparison

After: the Wilcoxon test

Change 405: replace | old pp. [7] | new pp. [7]

Before (including deletions): Human-Verified Net-Level Ground Truth (31 real images, detected wires). Primary metric is micro-averaged (pair-pooled) component-pair

After: per-image

Change 406: replace | old pp. [7] | new pp. [7]

Before (including deletions): macro (mean per-image) F1 is shown for reference.

After: W/T/L counts per-image wins, ties and losses from our side.

Change 407: replace | old pp. [7] | new pp. [7]

Before (including deletions): Prec Rec

After: P R

Change 408: replace | old pp. [7] | new pp. [7]

Before (including deletions): Deterministic

After: Δ [95% CI] Holm p (boot / perm / Wilcoxon) W/T/L Endpoint-graph

Change 409: delete | old pp. [7] | new pp. []

Before (including deletions): (ours and ablations)

After: (none)

Change 410: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: — — —

Change 411: delete | old pp. [7] | new pp. []

Before (including deletions): (prior)

After: (none)

Change 412: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: +0.061 [+0.025, +0.100] <0.001 / 0.001 / 0.009 11/18/2

Change 413: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: +0.075 [+0.038, +0.113] <0.001 / 0.002 / 0.005 14/16/1

Change 414: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: +0.103 [+0.065, +0.142] <0.001 / <0.001 / 0.002 17/14/0 Baselines on the same detected wires

Change 415: replace | old pp. [7] | new pp. [7]

Before (including deletions): Classical deterministic baselines (identical detected wires)

After: +0.224 [+0.166, +0.281] <0.001 / <0.001 / <0.001 25/6/0

Change 416: replace | old pp. [7] | new pp. [7]

Before (including deletions): Connected-components

After: +0.086 [+0.045, +0.127] <0.001 / 0.002 / 0.030 19/8/4 Connected components

Change 417: replace | old pp. [7] | new pp. [7]

Before (including deletions): Reference (non-deterministic)

After: +0.267 [+0.136, +0.409] <0.001

Change 418: replace | old pp. [7] | new pp. [7]

Before (including deletions): oracle wires

After: <0.001 / <0.001 21/8/2 References Ours with annotated wires 0.890 0.913 0.868 0.916 — — —

Change 419: replace | old pp. [7] | new pp. [7]

Before (including deletions): Scale-rel. + completion, perfect wires 0.890 0.913 0.868 0.916

After: $-0.033 [-0.078, +0.008]$ 0.12 / 0.15 / 0.030 7/11/13 on these 31 images, which favors them. We also tried to run five public image-to-netlist systems

Change 420: replace | old pp. [7] | new pp. [7]

Before (including deletions): authored synthetic suite (Table 3), where

After: benchmark. The printed-schematic interpreter of Kelly and Cole [4] runs, but on hand-drawn scans it over-segments wires into spurious components and recovers almost no connectivity. Netlistify [28] reproduces on its own printed AMS test data, but its released model lacks

Change 421: replace | old pp. [7] | new pp. [7]

Before (including deletions): scale- relative completion

After: diode, inductor and IC classes in our circuits, and its connectivity model is trained on printed thin-line drawings. Masala-CHAI [3] delegates connectivity to a proprietary LLM behind a paid API, which our VLM reference (Section V-I) covers. The code archive of SINA [1] could not be extracted, and CircuitNet [29] ships notebooks without trained weights. The CCL and Hough baselines therefore stand in for these approaches on identical inputs. Replacing the detected wires with the annotated wires, with boxes held fixed, changes micro-F1 from 0.8903 to 0.8898 and macro-F1 from 0.901 to 0.916. On this benchmark, the extracted wires are as good as the annotated ones for joining. The

Change 422: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: scores alike on both image modalities: micro- F1 is 0.883 on the 15 photographs and 0.895 on the 16 stroke maps. D. INDEPENDENT HELD-OUT BENCHMARK The 31-image benchmark has three weaknesses: it is small, its labels were bootstrapped from a graph join similar to ours, and our configuration was chosen on it. We therefore built a second benchmark whose labels come from the CGHD authors' own annotations and whose images were not used to develop or tune the join. The extractor configuration was selected earlier on the wire benchmark, whose resized copies include 25 of the held-out images. Excluding those 25 images (139 remain) gives micro-F1 0.709, and excluding every photograph of their drawings (123 remain) gives 0.704; in both cases our join still

Change 423: replace | old pp. [7] | new pp. [7]

Before (including deletions): so

After: with every margin still significant. Reference construction. Version 12 of CGHD [30] provides binary stroke-segmentation maps and symbol instance polygons for a subset of its images. Extending

Change 424: replace | old pp. [7] | new pp. [7]

Before (including deletions): result is not an artifact of how

After: dataset's own wire-generation tooling, we subtract

Change 425: replace | old pp. [7] | new pp. [7]

Before (including deletions): net-level labels were bootstrapped. The dominant connectivity recipe in prior image-to-netlist systems erases component boxes

After: dilated symbol polygons from the stroke map

Change 426: replace | old pp. [7] | new pp. [7]

Before (including deletions): reads nets off

After: take

Change 427: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: remaining

Change 428: replace | old pp. [7] | new pp. [7, 8]

Before (including deletions): components, or contours,

After: components as conductors. A conductor connects ev- VOLUME 11, 2023 7 0.6 0.8 1.0 Wire detection F1 (134 images) Triangle (skeleton) Otsu adaptive Gaussian (skeleton) Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola, anchor 12px Sauvola, anchor 16px, 12°/8px (deployed) Sauvola, anchor 16px, 10°/18px (best) 0.758 0.789 0.845 0.949 0.950 0.959 0.973 0.973 0.976 FIGURE 6. Wire detection F1 across thresholding methods and pipeline variants (134 images, annotated component boxes). The green bar is the deployed extractor (12■/8 px deduplication, F1 0.973), used in all other experiments; the top bar is the best

Change 429: replace | old pp. [7] | new pp. [8]

Before (including deletions): the residual wire pix- els. It is used by SINA [4], the concurrent contour-based node detector of Peker et al. [6], AMSnet [14],

After: 36 variants (10■/18 px, F1 0.976). The adaptive Gaussian

Change 430: replace | old pp. [7] | new pp. [8]

Before (including deletions): the Bayer instance-segmentation pipeline [7],

After: Triangle bars use skeleton extraction (F1 = 0.845

Change 431: replace | old pp. [7] | new pp. [8]

Before (including deletions): was designed for clean printed schematics. On hand-drawn photographs it has no precise operating point: pencil-stroke fragmentation splits each net into many components (F1 < 0.25 at sensible morphological gap-bridging), and only pathological closing that effectively merges by proximity lifts recall, at a steep precision cost. For a controlled comparison that isolates the join algorithm from binarization quality, we run connected- component labeling on our own detected wire segments, iden-

After: 0.758).

Change 432: replace | old pp. [7] | new pp. [8]

Before (including deletions): connectivity

After: component-pair

Change 433: delete | old pp. [7] | new pp. []

Before (including deletions): net-level

After: (none)

Change 434: replace | old pp. [7] | new pp. [8]

Before (including deletions): (component-pair metric). Our scale-relative completion join leads all deterministic strategies

After: with annotated component boxes

Change 435: delete | old pp. [7] | new pp. []

Before (including deletions): the classical Hough and connected-component baselines (run on identical

After: (none)

Change 436: replace | old pp. [7] | new pp. [8]

Before (including deletions): wires). The dashed line marks a strong VLM (Claude Opus 4.8) evaluated on the same 31 images, from which our join is statistically indistinguishable (paired micro-F1 difference +0.033, 95% bootstrap CI [-0.009, +0.078]) while being deterministic, structurally valid by construction, and orders of magnitude cheaper.

After: wires.

Change 437: replace | old pp. [7] | new pp. [8]

Before (including deletions): Cls. tical

After: Cls; the dashed line marks the VLM reference (Claude Opus 4.8). ery polygon it touches, except text; junction polygons merge the conductors they touch. At a crossover polygon, the arms are paired straight through by their local direction (180■± 45■), and a crossover that cannot be paired is flagged as ambiguous. Composite outlines (transformers, optocouplers and

relays that contain sub-parts) group their parts but take no contacts. Of 257 images, 179 pass all quality checks. Of the other 78, 16 contain no pair of electrical components and 62 fail at least one check: an ambiguous crossover (29 images), an isolated part (17) or an over-connected part (28), with 10 images failing more than one. Validation. On the 17 images shared with the human-verified benchmark, the reference agrees with the human nets at micro-F1 0.939 (P 0.993, R 0.891; exact on 11 of 17; 0.964 on the 15 clean images). We inspected every disagreement. Most are pairs the reference misses because of stroke gaps or ambiguous crossings, and seven follow from a convention: the human nets bridge switches. In five pairs, TABLE 6. Independent held-out benchmark (164 CGHD photographs, 24 drafters, 3874 reference pairs; annotated component boxes). Δ is ours minus the comparator in micro-F1, with 95% paired bootstrap CI. Every difference has Holm-adjusted $p = 0.0006$, the resolution floor of the tests. Method F1 P R Δ [95% CI] W/T/L Ours 0.711 0.766 0.664 – – Rescue + compl. 0.659 0.622 0.699 +0.053 [+0.033, +0.074] 85/44/35 Scale-rel. base 0.595 0.785 0.479 +0.117 [+0.097, +0.138] 112/46/6 Rescue base 0.592 0.631 0.558 +0.119 [+0.097, +0.142] 113/31/20 Radius union-find 0.518 0.478 0.565 +0.194 [+0.159, +0.229] 136/13/15 Hough 0.480 0.328 0.895 +0.231 [+0.168, +0.282] 90/22/52 Conn. comp. 0.594 0.798 0.473 +0.117 [+0.083, +0.156] 117/12/35 the reference was right and the human labels wrong. The reference mainly under-connects, so it understates the precision of every method. We fixed the quality filter and a junction-handling rule after inspecting these 17 overlaps. Protocol. The held-out set contains the 164 clean images outside the 31, with 3874 reference pairs from 24 drafters (all except drafter_0, which the detector's training also excluded). The

Change 438: replace | old pp. [7] | new pp. [8]

Before (including deletions): to our join,

After: is the CGHD photograph with its aspect ratio preserved

Change 439: replace | old pp. [7] | new pp. [8]

Before (including deletions): best configuration

After: long side resized to 1024 px. Component boxes are the CGHD symbol annotations, the same annotations the reference uses. We chose this resolution on wire-level labels of the 52 wire-benchmark overlaps, before computing any join score. Unlike the 31-image benchmark, which kept only circuits whose parts all belong to the scored device types, the held-out set also contains parts outside the scored types, most often switches and potentiometers (30 images each), speakers (21), transformers (16), thyristors (13) and logic gates (10); 94 of the 164 images contain at least one. These parts take part in the topology but are not scored. The detector merges potentiometers, photoresistors and phototransistors into resistor and transistor (Table 1), but the held-out scoring uses the CGHD annotation classes, in which they fall outside the scored types. Results. Table 6 gives the results. Our join

Change 440: replace | old pp. [7] | new pp. [8]

Before (including deletions): only micro-F1 = 0.624 (recall collapses to 0.46:

After: micro-F1 0.711 [0.671, 0.748] (P 0.766, R 0.664; macro-F1 0.724) and beats every comparator, with

Change 441: replace | old pp. [7] | new pp. [8]

Before (including deletions): blobs short or drop most nets), versus 0.890 for

After: same ordering of

Change 442: replace | old pp. [7] | new pp. [8]

Before (including deletions): join (Table 4). A stronger classical alternative, Hough-transform line detection [19] followed by proximity linking, fares better, since line fitting denoises

After: variants as on

Change 443: replace | old pp. [7] | new pp. [8]

Before (including deletions): fragmented strokes,

After: human-verified set. Hough linking keeps recall high (0.895) but merges many nets (precision 0.328); the best Hough configuration tuned on this set itself reaches 0.509. The result holds under two changes to the evaluation set: excluding all other photographs of the 31 benchmark drawings (155 images) gives 0.707,

Change 444: replace | old pp. [7] | new pp. [8]

Before (including deletions): reaches

After: scoring all 257 images including the unclean ones gives 0.626. Across the 24 drafters, our

Change 445: replace | old pp. [7] | new pp. [8]

Before (including deletions): = 0.805 at its best-tuned linking tolerance, but still trails

After: ranges from 0.465 to 0.917 (median 0.709, interquartile range 0.658–0.770), and

Change 446: delete | old pp. [7] | new pp. []

Before (including deletions): by 0.085 at markedly lower precision. Each baseline

After: (none)

Change 447: replace | old pp. [7] | new pp. [8]

Before (including deletions): reported at its

After: the

Change 448: replace | old pp. [7] | new pp. [8]

Before (including deletions): configuration over a tolerance sweep, so these

After: of the evaluated methods for 15 drafters. Scoring conventions. The primary scoring follows the human-verified benchmark: it scores only the electrical de- vice types, treats switches as components that

Change 449: replace | old pp. [7] | new pp. [8]

Before (including deletions): upper bounds on

After: not scored, and lets separate ground symbols join nodes only through drawn wires. We rescored

Change 450: replace | old pp. [7] | new pp. [8]

Before (including deletions): classical recipes. Explicit typed-endpoint reasoning with bounded completion thus outperforms both blob/contour connectivity

After: same predictions under three alternatives, applied symmetrically to reference

Change 451: replace | old pp. [7] | new pp. [8]

Before (including deletions): line-

After: predic- tions. Adding switches, potentiometers, photoresistors, fuses, 8

Change 452: replace | old pp. [7, 8] | new pp. [9]

Before (including deletions): 7 primitive proximity tracing.

After: lamps, varistors, crystals, speakers and microphones to the scored set (577 more reference pairs in 76 images) gives 0.709, with a margin of +0.055 [+0.036, +0.077] over the rescue-graph completion. Treating every switch as closed gives 0.712 (+0.052). Merging all ground symbols into one node, and all negative-supply symbols into another, adds 778 pairs in 43 images and gives 0.679 (+0.035 [+0.012, +0.059]); the merged ground net concentrates many pairs, so one missed ground attachment costs several. With all three changes combined, our join scores 0.676 and leads every baseline, with Holm-adjusted $p \leq 0.009$. Audit.

Change 453: replace | old pp. [8] | new pp. [9]

Before (including deletions): additionally attempted

After: drew 40 held-out images at random from the 164 and audited their derived nets. A human annotator checked 23 of them in the verification tool used for the human- verified benchmark, correcting any errors: the first 22 in the order the tool presents them, which runs from fewer

Change 454: replace | old pp. [8] | new pp. [9]

Before (including deletions): run five public image-to- netlist systems end-to-end

After: more components, and C64_D1_P3, which the annotator checked after the model pre-screen flagged it. The human-checked images are therefore mostly the smaller circuits (median 6 electrical components, against 9 for the 37 scored images and 10 for all 40 draws). A vision-language model used as an annotation aid (Claude Opus 5.5) pre-screened the 18

images outside those first 22. It saw only the image with numbered electrical-component boxes, never the derived nets, and traced each circuit. Its trace matched the derived nets exactly

Change 455: replace | old pp. [8] | new pp. [9]

Before (including deletions): our hand-drawn benchmark,

After: 11 images

Change 456: replace | old pp. [8] | new pp. [9]

Before (including deletions): none could be fairly applied.

After: differed on 4.

Change 457: replace | old pp. [8] | new pp. [9]

Before (including deletions): classical printed- schematic interpreter

After: human annotator adjudicated one

Change 458: replace | old pp. [8] | new pp. [9]

Before (including deletions): Kelly

After: the four (C64_D1_P3). For the other three, the model set the final nets after re-examining the disputed contacts; that step was not blind,

Change 459: delete | old pp. [8] | new pp. []

Before (including deletions): Cole [3] installs and runs, but on hand-drawn scans it over-segments wires into spurious components and recovers almost

After: (none)

Change 460: replace | old pp. [8] | new pp. [9]

Before (including deletions): connectivity, a domain- transfer failure. Netlistify [26] reproduces on

After: human checked

Change 461: replace | old pp. [8] | new pp. [9]

Before (including deletions): own printed AMS test data,

After: result. The model did not trace the three densest images (21– 30 electrical components),

Change 462: replace | old pp. [8] | new pp. [9]

Before (including deletions): ran end-to-end

After: excluded as too dense to verify reliably. The 37 scored images therefore consist of 23 human-checked, 11 model-matched and 3 model-adjudicated images. Over these 37, the reference scores micro-F1 0.988 against the final nets (precision 1.000, recall 0.975; 853 pairs) and is exact on 32 images. The 11 model-matched images agree

Change 463: replace | old pp. [8] | new pp. [9]

Before (including deletions): released model, but its active model lacks

After: reference by construction, so

Change 464: replace | old pp. [8] | new pp. [9]

Before (including deletions): diode, inductor,

After: direct evidence is the human-checked subset: on those 23 images the reference scores 0.988 (precision 1.000)

Change 465: replace | old pp. [8] | new pp. [9]

Before (including deletions): IC classes our circuits contain

After: is exact on 21. Our join scores 0.764 against the final nets of the 37 images

Change 466: replace | old pp. [8] | new pp. [9]

Before (including deletions): its connectivity model is trained on printed thin-line appearance, so it cannot represent hand-drawn discrete circuits. Masala-CHAI [5] delegates connectivity to

After: leads the rescue-graph completion by +0.055 [+0.026, +0.084], connected components by +0.116 and Hough linking by +0.219, with every CI excluding zero. Checking one image by hand took

Change 467: replace | old pp. [8] | new pp. [9]

Before (including deletions): proprietary LLM behind a paid API, equivalent in spirit to our VLM experiment in Section IV-F,

After: median of 85 s (mean 164 s) over 22 timed saves; the annotator verified proposed nets

Change 468: replace | old pp. [8] | new pp. [9]

Before (including deletions): a deterministic algorithm. SINA's [4] anonymized code archive

After: labeling from scratch. Gap to the human-verified benchmark. Part of the drop from 0.890 to 0.711 comes from the input images. The held-out photographs belong to the CGHD subset for which the dataset provides stroke maps, and for this subset the 704×704 benchmark copies are mostly those stroke maps (16 of the 17 overlaps). Many of these photographs show grid paper or cluttered backgrounds, which the 31-image copies rarely contain; on them, wire F1 falls to about 0.80–0.82, against 0.98 on the benchmark copies. On the 17 overlapping images, with our annotated boxes and the human nets, our join scores TABLE 7. Micro-F1 by number of electrical components per image. The VLM

Change 469: replace | old pp. [8] | new pp. [9]

Before (including deletions): unextractable

After: run on the human-verified benchmark only. Components Images Ours Rescue + compl. Hough VLM Human-verified benchmark (31 images) ≤ 5 15 0.940 0.933 0.892 0.982 6–9 4 0.846 0.817 0.843 0.902 ≥ 10 12 0.886 0.808 0.779 0.912
Held-out benchmark (164 images) ≤ 5 41 0.845 0.856 0.835 – 6–9 41 0.779 0.756 0.701 – 10–19 60 0.694 0.656 0.593 – ≥ 20 22 0.693 0.608 0.347 – 0.896 on the benchmark copies

Change 470: replace | old pp. [8] | new pp. [9]

Before (including deletions): CircuitNet [27] ships only Co-lab notebooks without released weights. This survey is itself informative: no prior system provides a deterministic, hand-drawn-ready connectivity module,

After: 0.728 on the photographs at long side 1024. Re-weighting the held-out images to the circuit-size distribution of the 31 raises our score by about 0.01. The stored held-out predictions for the same 17 photographs, which use the CGHD symbol boxes, score 0.714 against the human nets and 0.723 against the reference,

Change 471: replace | old pp. [8] | new pp. [9]

Before (including deletions): controlled CCL

After: choice of labels changes the score by about 0.01; on these 17 photographs Hough linking scores higher than our join (0.788 against the human nets, 0.738 against the reference). Input quality explains part of the gap, but not all of it: with the CGHD stroke maps as input instead of the photographs, our held-out score rises only to 0.775. The residual is consistent with some optimism in the 31-image score, on which our configuration was selected. Across both benchmarks, our join ranks first and the endpoint-graph variants keep their order; the classical baselines do not keep theirs (Hough ranks below connected components on the held-out set). E. ROBUSTNESS 1) Circuit complexity Table 7 splits both benchmarks by the number of electrical components. On the 31 images, F1 falls with complexity for every method, including the VLM: across images, our F1 correlates with component count at Spearman $\rho = -0.53$ ($p = 0.002$). Our recall drop from the ≤ 5 to the ≥ 10 stratum, -0.068 $[-0.18, +0.05]$ (permutation $p = 0.32$), is not significant. On images with at least ten components, our join beats the rescue-graph completion, the scale-relative base

Change 472: replace | old pp. [8] | new pp. [9]

Before (including deletions): baselines above, both run

After: by +0.078 to +0.106, with every CI excluding zero, and trails the VLM by -0.026 $[-0.087, +0.030]$. On the held-out benchmark, micro-F1 falls from 0.845 for at most five components to 0.693 for 20 or more, while Hough falls to 0.347. Our recall is 0.773, 0.741, 0.664 and 0.623 in the ≤ 5 , 6–9, 10–19 and ≥ 20 strata. Connected components is the only method whose F1 is higher

Change 473: delete | old pp. [8] | new pp. []

Before (including deletions): our identical detected wires, are

After: (none)

Change 474: replace | old pp. [8] | new pp. [9]

Before (including deletions): fair points

After: largest circuits than on the smallest (0.519 to 0.655), because its low recall rises with circuit size. 2) Component-size variation within an image A single scale s per image could fail when component sizes vary within a drawing. On the 31 images, the overall dis- persion

Change 475: replace | old pp. [8] | new pp. [9, 10]

Before (including deletions): comparison, and

After: component sizes is unrelated to F1 ($p = +0.18$, VOLUME 11, 2023 9 $p = 0.33$). The ratio of the largest to the smallest electrical- component diagonal correlates with F1 at $p = -0.60$, but it also grows with component count ($p = 0.62$); controlling for count, the partial correlation is -0.41 ($p = 0.024$) for

Change 476: replace | old pp. [8] | new pp. [10]

Before (including deletions): exceeds both. Running

After: and -0.34 ($p = 0.064$) for

Change 477: replace | old pp. [8] | new pp. [10]

Before (including deletions): best join

After: VLM. The size of the largest component relative to s has partial correlation near zero with F1. The three images that combine an IC with small discrete parts score 0.854 against 0.898

Change 478: replace | old pp. [8] | new pp. [10]

Before (including deletions): dataset's perfect

After: other 28. 3) Controlled rescaling We resampled all 31 images by factors from 0.35 to 3 and re-projected the annotated boxes and wires (Fig. 8). With annotated wires, our join stays within ± 0.005 of its native score from $0.5\times$ to $3\times$, with every CI including zero. It drops only at $0.35\times$ (-0.035 [-0.061 , -0.005]), where the lower pixel clamps bind for every image. With annotated wires, radius union-find with its fixed 30 px radius falls from 0.690 at native scale to 0.429 at $2\times$ and 0.153 at $3\times$. With wires re- extracted from each rescaled image, the full pipeline scores 0.580, 0.690, 0.811, 0.890, 0.877, 0.866 and 0.866 at 0.35, 0.5, 0.75, 1, 1.5, 2 and $3\times$. Upscaling costs at most 0.024 (CIs include zero). The loss under downscaling comes from

Change 479: replace | old pp. [8] | new pp. [10]

Before (including deletions): labels rather

After: extraction, whose Sauvola window, minimum compo- nent area and occlusion margins are set in pixels: at $0.35\times$ the extractor finds 241 of 675 annotated segments. Completion recovers much of that loss; the scale-relative base alone scores 0.21 at $0.35\times$. To separate the contribution of the scale-relative tolerances, we ran the same algorithm with fixed pixel tolerances (τ_{pin} , τ_{join} , τ of 30, 14 and 10 px, completion reach 120 px). It stays within 0.016 of our join between $0.5\times$ and $2\times$. The scale-relative version wins only where the fixed values are most mismatched: by $+0.015$ with annotated wires at $0.5\times$ and by $+0.033$ in the full pipeline at $3\times$, both with CIs excluding zero. The fixed-pixel variant is ahead at $1.5\times$ with annotated wires (-0.012 [-0.025 , -0.001]) and at $0.75\times$ in the full pipeline (-0.016 [-0.034 , -0.001]). Most of the scale robustness therefore comes from component-relative endpoint assignment and completion; the scalar tolerances $\tau = k \cdot s$ contribute a smaller share. The clamps are tuned to native resolution: $k \cdot s$ falls outside a clamp for 39% of images at native scale, and removing the clamps costs 0.023 in the full pipeline at native scale. 4) Synthetic mixed-size circuits We redrew one component per synthetic circuit at 0.5, 2 or 3 times its size, keeping only geometrically clean variants, and applied the L1–L4 error model. Enlarging one component costs our join about 0.005 in mean F1, and it stays about 0.03 above the fixed-pixel variant. Shrinking one component costs 0.012. The failure case is a Wheatstone bridge with one 40×15 px arm: in all five variants, the scale-relative methods and radius union-find add one spurious pair even on clean wires, while the fixed-pixel graphs stay exact. A component much smaller

Change 480: insert | old pp. [] | new pp. [10, 11]

Before (including deletions): (none)

After: the median is thus the weak point of a median-derived tolerance. 0.0 0.2 0.4 0.6 0.8 1.0 Connectivity micro-F1 (a) Join only: annotated wires 0.35 0.5 0.75 1 1.5 2 3 Image scale factor f (log scale; native = 1) 0.0 0.2 0.4 0.6 0.8 1.0 Connectivity micro-F1 (b) Full: wires re-extracted 95% bootstrap CI Scale-rel. graph + completion Fixed-px graph + completion Scale-rel. graph (base) Radius union-find (30 px) FIGURE 8. Component-pair micro-F1 on the 31 images resampled by factor f (native $f = 1$). (a) Join only, with annotated wires rescaled with the image. (b) Full pipeline, with wires re-extracted from each rescaled image. Components are annotated boxes in both panels; shaded bands are 95% bootstrap CIs for our join. TABLE 8. Edge-rule ablation on the 31-image benchmark, for the base graph and the full pipeline with completion. Base graph Full pipeline Configuration F1 TP/FP/FN F1 TP/FP/FN All edge types (ours) 0.816 352/27/132 0.890 418/37/66 No T-junctions 0.816 352/27/132 0.890 418/37/66 No rail taps 0.816 352/27/132 0.890 418/37/66 No T-junctions, no rail taps 0.816 352/27/132 0.890 418/37/66 No directional preference 0.816 352/27/132 0.890 418/37/66 Fixed-pixel base tolerances 0.820 356/28/128 0.890 418/37/66 F. ABLATIONS Table 8 removes individual edge rules from the base graph, with and without completion. Disabling T-junctions, rail taps or directional preference leaves the aggregate base-graph counts unchanged, so these rules give no measurable gain on this benchmark. Fixed-pixel base tolerances raise base micro-F1 from 0.816 to 0.820. With completion, every row ties at TP/FP/FN 418/37/66. The rescaling and mixed-size experiments (Section V-E) test the tolerance design under conditions that this benchmark does not cover. Table 9 tests the remaining mechanisms. Removing only the occlusion step, with cropping and all extractor settings held fixed, exposes component pixels to the extractor, which then returns 466 wire segments instead of 666; micro-F1 falls from 0.890 to 0.336 (P 0.798, R 0.213), and predic- 10 VOLUME 11, 2023 TABLE 9. Mechanism ablations on the 31-image benchmark. “Changed” counts images whose predicted pairs differ from the reference arm. Arm TP/FP/FN F1 F1macro Changed Occlusion on (ours) 418/37/66 0.890 0.901 – Occlusion off 103/26/381 0.336 0.294 31/31 Guard on (ours) 418/37/66 0.890 0.901 – Guard off 418/37/66 0.890 0.901 0/31 Witness required 413/37/71 0.884 0.907 6/31 TABLE 10. End-to-end joining with

Change 481: replace | old pp. [8] | new pp. [11]

Before (including deletions): wires leaves

After: components on the 31-image benchmark. The lower block replaces annotated boxes with detector output one factor at a time (confidence 0.5); Δ is the change from the previous row. Components F1 Prec Rec TP/FP/FN Annotated boxes 0.890 0.919 0.864 418/37/66 Detector, conf 0.25 0.613 0.575 0.655 317/234/167 Detector, conf 0.35 0.616 0.582 0.655 317/228/167 Detector, conf 0.50 0.627 0.603 0.653 316/208/168 Sequential step F1 Δ TP/FP/FN Drop missed components 0.759 –0.131 329/54/155 Add spurious detections 0.687 –0.072 322/131/162 Use detected geometry 0.644 –0.043 329/208/155 Use detected classes 0.627 –0.017 316/208/168 tions change on all 31 images. The shared-component guard fires on 18 of 31 images and rejects 38 candidate merges. These merges would join two pins of the same component, which the component-pair metric cannot see, so the scores are unchanged. Requiring a wire witness for every completion edge changes six images and lowers

Change 482: replace | old pp. [8] | new pp. [11]

Before (including deletions): unchanged at 0.890,

After: to 0.884

Change 483: replace | old pp. [8] | new pp. [11]

Before (including deletions): only

After: to 0.907. The rescaling experiment adds two further arms, fixed-pixel tolerances and unclamped tolerances (Section V-E), and Section V-G decomposes the loss

Change 484: replace | old pp. [8] | new pp. [11]

Before (including deletions): 0.901 to 0.916 (Table 4), so

After: detected components. G. END-TO-END WITH DETECTED COMPONENTS We replaced

Change 485: replace | old pp. [8] | new pp. [11]

Before (including deletions): wire detector accounts for essentially zero pair-weighted F1. The residual error is intrinsic join ambiguity on hand-drawn geometry, not wire detection, consistent

After: annotated boxes

Change 486: replace | old pp. [8] | new pp. [11]

Before (including deletions): 0.976 wire-detection

After: outputs of our trained component detector (Section III-B). The detector runs on the CGHD original, corrected for its Exchangeable Image File Format (EXIF) orientation tag, at its training size (long side 1024 px), and its boxes are mapped into the benchmark frame. We use the deployment confidence threshold of 0.5 without tuning. On the electrical components, detection pre- cision, recall and

Change 487: replace | old pp. [8] | new pp. [11]

Before (including deletions): reported above (Table 2).

After: are 0.886, 0.858 and 0.872, and the same when the class must also match. As a check of the scoring path, feeding the annotated boxes through the same mapping and matching code reproduces 418/37/66. Table 10 gives the results. End-to-end micro-F1 is 0.627 [0.519, 0.731] (macro-F1 0.656), a change of -0.263 [-0.366 , -0.170] from annotated boxes; the lower thresh- olds give 0.613 and 0.616. Replacing the annotated boxes one factor at a time attributes -0.131 to missed electrical components, -0.072 to spurious detections, -0.043 to box localization and -0.017 to classification; with a sequential decomposition, the shares depend on the order of the steps. Missed components cost the most because every pair in- volving a missed component becomes a false negative.

Change 488: replace | old pp. [8] | new pp. [11]

Before (including deletions): join degrades only gently with circuit size, with per-image F1 falling from 0.94 on circuits with ≤ 6 components

After: detector is sensitive

Change 489: replace | old pp. [8] | new pp. [11]

Before (including deletions): 0.86 above that and only weak correlation $r = -0.19$, and remains high-precision throughout, in contrast

After: orientation (its training used rotation augmentation of at most $\pm 10^\circ$). On the 14 images whose benchmark copy is re-oriented relative

Change 490: replace | old pp. [8] | new pp. [11]

Before (including deletions): VLM, whose precision collapses

After: CGHD file, end- to-end micro-F1 is 0.491, against 0.739

Change 491: replace | old pp. [8] | new pp. [11]

Before (including deletions): densest circuits (Section IV-F). E. PER-CIRCUIT ANALYSIS Table

After: other 17. H. CROSSOVERS Edge types 2, 4 and

Change 492: replace | old pp. [8] | new pp. [11]

Before (including deletions): shows

After: do not consult crossover labels, so a missed crossover could merge two independent wires. The 31 images contain 13 annotated crossovers in eight images. With annotated boxes, recall on these eight images is 0.834, against 0.884 on

Change 493: replace | old pp. [8] | new pp. [11]

Before (including deletions): scale-relative completion join's perfor- mance across eight representative circuits

After: 23 without a crossover ($p = 0.51$). In C242_D1_P1, suppressing the two endpoint–endpoint edges adjacent to crossings changes TP/FP/FN

Change 494: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: 27/4/0 to 19/0/8:

Change 495: replace | old pp. [8] | new pp. [11]

Before (including deletions): 15-circuit synthetic corpus at L4 (extreme error injection, 16 seeds). TABLE 5. Scale-Relative Completion Join Performance per Circuit at L4 (16 seeds) Circuit Comps F1 Prec Sim. acc. Parallel R–R 3 1.00 1.00 25% R–R divider 3 0.99 1.00 69% RL series 3 0.99 1.00 81% Diode–R loop 3 0.99 1.00 69% Dual R loops 4 0.93 0.90 12% 4-component

loop 4 0.94 0.91 56% Wheatstone bridge

After: edges that create the false pairs also carry true connections. Deleting the annotated crossover boxes before joining changes predictions on

Change 496: replace | old pp. [8] | new pp. [11]

Before (including deletions): 0.82 0.85 6% 6-component ring 6 0.95 0.92 50% Precision remains high (≥ 0.85) across all circuits even at L4: errors cause fragmentation (dropped connections) rather than shorts (false merges). The Wheatstone bridge is

After: of

Change 497: replace | old pp. [8] | new pp. [11]

Before (including deletions): hard- est topology ($F1 = 0.82$): its balanced arms make

After: 8 crossover images and raises micro-F1 by 0.010, so on annotated boxes

Change 498: replace | old pp. [8] | new pp. [11]

Before (including deletions): inferred node structure sensitive to a single mis-placed connection, and simulation accuracy falls to 6%. F. VLM CONNECTIVITY EXPERIMENT To quantify the separation argued in Section II, that connec- tivity is better solved by an explicit geometric model than left to a VLM, we ran a first-party experiment with a strong general VLM (Claude Opus 4.8). To make the comparison fair, the VLM is given exactly what

After: crossover labels do not help

Change 499: replace | old pp. [8] | new pp. [11]

Before (including deletions): receives:

After: overall. In

Change 500: replace | old pp. [8] | new pp. [11]

Before (including deletions): scanned

After: end-to-end run, the detector finds all 13 crossovers and labels each as a crossover at every tested threshold, so none of the false pairs in Table 10 comes from a missed or misclassified crossover; removing the detected crossovers lowers end-to-end micro- F1 by 0.019. These images were probably in the detector’s training split, so this is not a held-out estimate of crossover recall, and thirteen instances are too few to test the 70.7% validation recall. I. VLM REFERENCE We ran Claude Opus 4.8 as a reference system. It receives the annotated boxes of the electrical components, drawn on the

Change 501: replace | old pp. [8] | new pp. [11]

Before (including deletions): with

After: and numbered without class labels. Our join receives

Change 502: delete | old pp. [8] | new pp. []

Before (including deletions): ground-truth component

After: (none)

Change 503: replace | old pp. [8] | new pp. [11]

Before (including deletions): labelled by index,

After: with their classes, plus the annotated junc- tion, terminal, ground

Change 504: replace | old pp. [8] | new pp. [11]

Before (including deletions): is asked to return, as JSON,

After: crossover boxes. Neither method receives annotated wires. The VLM was run on

Change 505: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: human- verified benchmark only. The model returns JSON

Change 506: replace | old pp. [8] | new pp. [11]

Before (including deletions): we

After: each image is a separate call with no shared context. We

Change 507: replace | old pp. [8] | new pp. [11]

Before (including deletions): the resulting component-pairs

After: its output

Change 508: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: component-pair

Change 509: replace | old pp. [8] | new pp. [11]

Before (including deletions): Each image is read by an independent, context-free model call (one per image), so no answer leaks between circuits, and the human, not the model, remains the verifier of record. On a clean synthetic control (15 authored circuits) the

After: The

Change 510: delete | old pp. [8] | new pp. []

Before (including deletions): scores $F1 = 0.99$, confirming it understands the task and symbol set. On all 31 real scans, evaluated against the human-verified net-level ground truth, it

After: (none)

Change 511: delete | old pp. [8] | new pp. []

Before (including deletions): =

After: (none)

Change 512: replace | old pp. [8] | new pp. [11]

Before (including deletions): = 0.97,

After: 0.970,

Change 513: replace | old pp. [8] | new pp. [11]

Before (including deletions): = 0.88;

After: 0.880;

Change 514: replace | old pp. [8] | new pp. [11]

Before (including deletions): = 0.949). The VLM thus edges our deterministic join on aggregate (0.923 vs 0.890 micro-F1), but the gap is not statistically significant: a paired bootstrap over the 31 images puts the difference at +0.033 with a 95% CI of $[-0.009, +0.078]$, which includes zero. Two further findings qualify the comparison. First, the VLM's accuracy degrades with circuit complexity: it

After: 0.949) and

Change 515: delete | old pp. [8] | new pp. []

Before (including deletions): ($F1 = 1.00$)

After: (none)

Change 516: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: 31 images, against 11 for our join. On pooled pairs, our join trails by -0.033 $[-0.078, +0.008]$, which is not significant (Holm-adjusted bootstrap $p = 0.12$, permutation $p = 0.15$; Table 5). Per image,

Change 517: replace | old pp. [8] | new pp. [11]

Before (including deletions): 31 circuits, including all

After: VLM is better: it wins 13 images, ties 11 and loses 7 (Wilcoxon $p = 0.030$). Its errors follow

Change 518: replace | old pp. [8] | new pp. [11, 12]

Before (including deletions): small 3–5-component cases, but

After: same complex-ity trend as ours (Table 7). The two systems differ in what they require: the VLM needs a call to a proprietary API for every image and returns no geometric trace of its decisions, whereas our join runs locally and deterministically without connectivity training data, and every recovered connection maps to a wire segment or a completion edge. VOLUME 11, 2023 11 TABLE 11. Micro-F1 of our join by CGHD drafter

Change 519: replace | old pp. [8] | new pp. [12]

Before (including deletions): largest boards it systematically omits connections,

After: 31-image benchmark (drafters

Change 520: replace | old pp. [8] | new pp. [12]

Before (including deletions): recall falling to 0.63–0.68 on

After: at least four images). Drafter Images F1 F1macro TP/FP/FN drafter_1 7 0.933 0.907 56/6/2 drafter_2 5 0.854 0.874 85/14/15 drafter_10 5 0.987 0.995 37/1/0 drafter_12 4 0.913 0.938 21/0/4 All 31 0.890 0.901 418/37/66 J. PER-DRAFTER RESULTS Table 11 groups

Change 521: replace | old pp. [8] | new pp. [12]

Before (including deletions): three densest boards, exactly where a netlist error is most costly; its precision otherwise stays high (0.97 micro), so its failure mode is dropped nets rather than shorts. The macro-F1 (0.949) is correspondingly inflated

After: 31-image results

Change 522: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: CGHD drafter, using

Change 523: replace | old pp. [8] | new pp. [12]

Before (including deletions): many easy small circuits, which is why we report the pair-weighted micro-F1 as primary. Second, the cost and form

After: drafter paths

Change 524: replace | old pp. [8] | new pp. [12]

Before (including deletions): output differ

After: published CGHD-1152 file index [15], which place every benchmark filename under exactly one drafter. The four drafters with at least four images range

Change 525: replace | old pp. [8] | new pp. [12]

Before (including deletions): the geometric pipeline: each image consumes on the order of 105 tokens, the result is free-form text rather than a structured netlist, and nothing in the model guarantees electrical validity (e.g. that a two-terminal component is not shorted). Our geometric join is statistically indistinguishable from the VLM on these circuits while being deterministic, two

After: micro-F1 0.854

Change 526: replace | old pp. [8] | new pp. [12]

Before (including deletions): three orders of magnitude cheaper, and structurally valid by construction: the self-loop guard guarantees that no two-terminal component is emitted shorted. We therefore treat the VLM as an upper reference for what a frontier model can do here, not as a connectivity module to ship: an explicit model reaches the same accuracy and also

After: 0.987. The held-out benchmark

Change 527: replace | old pp. [8] | new pp. [12]

Before (including deletions): determinism, low cost, and structural validity a deployed digitizer needs. V.

After: spread over all 24 drafters (Section V-D). VI.

Change 528: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: join with completion is the strongest deterministic method on both real benchmarks, and its margins are significant on both, which differ in labels and images and share part of their drafters. The held-out benchmark addresses the three main weaknesses of the human-verified set: its labels come from the dataset authors' annotations and not from a join, it covers 24 of the 25 drafters, and its images were not used to develop the join. Completion accounts for most of the gain over the base graphs, and on the high-precision scale- relative base it adds recall while keeping precision above 0.9 on the human-verified set. The experiments also locate the remaining errors. With annotated components on clean copies, extracted wires are as good as annotated ones. On difficult photographs, wire extraction limits accuracy; with detected components, missed detections do. Three changes follow from these results: an extractor whose pixel parameters follow the component scale, resampling inputs to a nominal component size, and a detector trained with full rotation augmentation. The VLM reference shows that a general

Change 529: replace | old pp. [8] | new pp. [12]

Before (including deletions): resolves ambiguity that radius-based methods cannot:

After: given component boxes can exceed a geometric join per image on this task. The geometric join remains useful where a local, deterministic and inspectable result is required. A. SCOPE AND LIMITATIONS Task and metric. The pipeline recovers structural connectivity. It does not read component values or device models, so its illustrative netlist export (SPICE syntax) uses placeholder values; simulation of real scans is not evaluated. The component-pair metric ignores terminal identity: a perfect score does not certify pin assignments, the exact net partition or the absence of shorts between pins of one component. The completion guard blocks new same-component merges but does not repair merges made

Change 530: replace | old pp. [8] | new pp. [12]

Before (including deletions): representing both

After: the base graph. Connectivity is read from drawn wires only: terminals and supply symbols are

Change 531: replace | old pp. [8, 9] | new pp. [12]

Before (including deletions): end- 8 VOLUME 11, 2023 points

After: ends, so identically labeled supply terminals (e.g., two "+5 V" points)

Change 532: replace | old pp. [9] | new pp. [12]

Before (including deletions): component pins as nodes with typed edges, it distinguishes

After: the implied source

Change 533: replace | old pp. [9] | new pp. [12]

Before (including deletions): wire that passes near a component (endpoint-wire-body edge)

After: pair of input terminals are not interpreted, which would require reading their labels. Table 12 lists the device groups: pin templates for transformers, optocouplers

Change 534: replace | old pp. [9] | new pp. [12]

Before (including deletions): one that connects to it (endpoint-pin edge with directional preference).

After: other complex devices exist but are not validated.

Change 535: replace | old pp. [9] | new pp. [12]

Before (including deletions): component-first assignment further disambiguates by using the geometric constraint that wires must reach a component's pin, not just its bounding box. Degree-budget completion addresses the fundamental limitation of detector noise: even a 97.6% accurate wire detector drops ~2% of wire endpoints, and those dropped endpoints are precisely the connections that matter most for simulation accuracy. The b-matching formulation bounds the damage from mis-detection (each floating pin gets at most one recovery edge), while the self-loop guard prevents the catastrophic short-circuit failures that plague unconstrained approaches. The human-verified net-level evaluation, while novel for hand-drawn circuits, currently covers 31 images; the 95% bootstrap confidence intervals on the per-strategy F1 are correspondingly wide (± 0.03 – 0.05) and overlap between adjacent strategies, so while the top ranking is consistent and corroborated by the independently authored synthetic suite, broader annotation is needed to separate the closely ranked configurations. The net-level labels were bootstrapped by a conservative endpoint-graph join over perfect wires and then human-corrected; we mitigate the residual concern that this favors graph-based methods by validating on synthetic ground truth that is authored from scratch, where the same ranking holds. The evaluation draws from a single hand-drawn corpus (CGHD-1152); generalization across other drawing styles, capture conditions, and printed schematics remains to be validated, as does extension to circuits containing devices outside the simulatable set (switches, transformers, etc.), which we exclude from the net-level benchmark. The VLM connectivity comparison uses a single frontier model (Claude Opus 4.8) on the same 31 images; evaluating additional models and prompting strategies is future work. Component detection, at $\text{mAP}@0.5 = 89.0\%$ and 70.7% recall on crossovers (Section III-B), also leaves room for improvement; moreover the

After: detector's

Change 536: replace | old pp. [9] | new pp. [12]

Before (including deletions): together,

After: together (Section III-B),

Change 537: replace | old pp. [9] | new pp. [12]

Before (including deletions): the detector localizes

After: with detected components

Change 538: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: are localized

Change 539: replace | old pp. [9] | new pp. [12]

Before (including deletions): cannot name them,

After: not named,

Change 540: replace | old pp. [9] | new pp. [12]

Before (including deletions): containing

After: that contains

Change 541: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: Data. All images come from one corpus. The human-verified nets were bootstrapped from a conservative endpoint-graph join and corrected by a single annotator, which may leave a bias toward graph-based methods; on the held-out benchmark, with labels from the dataset's own annotations, our join again ranks first. The inspection of the 17 overlaps found five human-label errors (four in C242_D1_P1); we left the released human-verified labels unchanged. Only 37 of the 164 held-out images were audited, and only 23 of them, mostly the smaller circuits, were checked by a human. The other 14 were settled with a blind model pre-screen, and for three of them the model set the final nets after re-examining the disputed contacts, without a human check. The three densest of the 40 draws were excluded. The reference mainly misses connections, and its quality filter and a junction rule were fixed after inspecting the 17 overlapping images. The benchmark copies of the 31 images are resized, partly re-oriented, and half stroke maps, which makes them easier than raw photographs of the same drawings. The Hough and CCL baselines keep their 31-image settings on the held-out photographs. Scale. The join tolerances derive from one median scale per image, and their clamps are tuned to native resolution. A component much smaller than the median can create a spurious pair, as in the synthetic Wheatstone case. The extractor's pixel-valued

parameters limit accuracy on downscaled inputs. Rescaling tests changes in zoom, not in drawing style, and uses annotated components. Detector. The detector was trained on CGHD images with an 85/15 random split, and no manifest of that split survives. The 31 benchmark drawings come from drafters eligible for training, and about 26 of them would be expected in the training split, so the end-to-end numbers are likely optimistic for detection.

Change 542: replace | old pp. [9] | new pp. [12]

Before (including deletions): avoids domain-shift artifacts from an outlier drawing style, but

After: (Section III-B)

Change 543: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: drawing

Change 544: replace | old pp. [9] | new pp. [12]

Before (including deletions): unmodelled.

After: unmodeled, and the detector is sensitive to orientation. Coverage.

Change 545: replace | old pp. [9] | new pp. [12]

Before (including deletions): pipeline recovers circuit topology,

After: human-verified benchmark contains 13 annotated crossovers, and we did

Change 546: replace | old pp. [9] | new pp. [12]

Before (including deletions): component values: simulation accuracy is reported

After: evaluate dense multi-track buses or multi-layer crossings. The VLM reference uses one model and one prompt, and was run

Change 547: replace | old pp. [9] | new pp. [12]

Before (including deletions): synthetic suite, where values are known by construction, and no netlist derived from a real scan is simulated here. Producing a numerically simulatable netlist from a scan additionally requires reading component values off the image, which we do not attempt; coupling the join to a value-recognition (OCR) stage, as in [3], [4], and validating end-to-end against a reference simulation, as in [6], is the natural completion of the pipeline. Finally, the

After: human-verified benchmark. The

Change 548: replace | old pp. [9] | new pp. [12]

Before (including deletions): a first-pass approximation of detector noise rather than a

After: authored and not

Change 549: delete | old pp. [9] | new pp. []

Before (including deletions): fit; with the new real-image net-level metric available, calibrating it

After: (none)

Change 550: delete | old pp. [9] | new pp. []

Before (including deletions): the

After: (none)

Change 551: replace | old pp. [9] | new pp. [12, 13]

Before (including deletions): failure distribution

After: errors. Expansion. The derived reference makes a larger human-verified benchmark affordable: its nets can seed the verification tool, where checking an image took a median of 85 s in our audit. We plan to sample new images stratified by component count, drafter, crossing and bus structure and capture conditions, have two annotators correct each image 12

VOLUME 11, 2023 TABLE 12. Connectivity evaluation, pin construction and export by device group. Device group Evaluated scope Pin construction Export R/C/L/D/Q, voltage sources and ICs Component-pair connectivity on both benchmarks Geometric templates; endpoint clustering for discrete and source types, generic layouts for ICs Placeholder values; ICs are skipped because no export model

Change 552: replace | old pp. [9] | new pp. [13]

Before (including deletions): a natural next step. VI.

After: implemented. Switches Scored in the extended held-out variant (Section V-D); not in the primary subset Two pins at the shorter OBB-edge midpoints; long-axis fallback A 0.001 Ω resistor if the two pins map to distinct nodes; no switch state is recovered. Transformers, thyristors, optocouplers and other complex devices Not evaluated Two-pin geometric guess for transformers; optocoupler template; generic fallback No export model; skipped. independently with adjudication of disagreements, and extend the evaluation to printed schematics and to corpora other than CGHD. VII.

Change 553: replace | old pp. [9] | new pp. [13]

Before (including deletions): present

After: presented

Change 554: delete | old pp. [9] | new pp. []

Before (including deletions): complete

After: (none)

Change 555: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: structural netlist recovery from

Change 556: replace | old pp. [9] | new pp. [13]

Before (including deletions): circuit digitization,

After: schematics,

Change 557: replace | old pp. [9] | new pp. [13]

Before (including deletions): occlusion-first

After: occlusion-first

Change 558: replace | old pp. [9] | new pp. [13]

Before (including deletions): extractor that yields an endpoint representation and

After: extractor,

Change 559: replace | old pp. [9] | new pp. [13]

Before (including deletions): two-stage endpoint-graph join, together with, to our knowledge, the first human-verified net-level connectivity benchmark for hand-drawn circuits. The

After: typed

Change 560: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: and degree-budget completion. With annotated components, the join reaches component-pair micro-F1 0.890 on 31 human-verified images and 0.711 on 164 held-out photographs

Change 561: replace | old pp. [9] | new pp. [13]

Before (including deletions): degree-budget completion is most robust under synthetic error injection (F1 = 0.95 at the highest severity vs. 0.36 for the radius baseline), and,

After: independently derived reference nets, and it beats every deterministic baseline on both with significant

Change 562: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: margins. Its accuracy holds from 0.5× to 3× image scale when wires are given, and it reaches 0.627 end to end

Change 563: replace | old pp. [9] | new pp. [13]

Before (including deletions): high-precision scale-relative base, attains connectivity micro-F1 = 0.890

After: trained detector. A VLM given the electrical-component boxes scores higher per image, but requires a proprietary model call per image. Better extraction

Change 564: replace | old pp. [9] | new pp. [13]

Before (including deletions): real images against human-verified ground truth, beating

After: difficult photographs and better component detection are

Change 565: replace | old pp. [9] | new pp. [13]

Before (including deletions): prior completion default (0.83) and every deterministic baseline, including the connected-component net-tracing recipe used by recent systems (0.62) and a Hough-plus-proximity line tracer (0.81), all on identical detected wires. Running the join on perfect wires does not raise micro-F1 at all, showing the join, not wire detection, is the bottleneck. A first-party experiment with a frontier VLM reaches 0.923 on the same images, a difference from our join that is not statistically significant (paired 95% CI [−0.009, +0.078]), but at two

After: most direct routes

Change 566: delete | old pp. [9] | new pp. []

Before (including deletions): three orders of magnitude

After: (none)

Change 567: replace | old pp. [9] | new pp. [13]

Before (including deletions): cost and without structurally valid output, supporting the explicit geometric design. Future work includes scaling the human-verified benchmark, calibrating the synthetic error model to the measured detector distribution, and extending coverage to devices outside the simulatable set.

After: end- to-end accuracy.

Change 568: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: models (Anthropic;

Change 569: replace | old pp. [9] | new pp. [13]

Before (including deletions): (Anthropic)

After: and Opus 5.5)

Change 570: delete | old pp. [9] | new pp. []

Before (including deletions): a

After: (none)

Change 571: replace | old pp. [9] | new pp. [13]

Before (including deletions): assistant

After: assistants

Change 572: replace | old pp. [9] | new pp. [13]

Before (including deletions): development

After: develop- ment

Change 573: replace | old pp. [9] | new pp. [13]

Before (including deletions): code.

After: code, and Claude Opus 5.5 served as the blind pre-screening aid in the held-out audit (Section V-D) and set the final nets of three audit images after re-examining the disputed contacts.

Change 574: replace | old pp. [9] | new pp. [13]

Before (including deletions): IV-F,

After: V-I,

Change 575: replace | old pp. [9] | new pp. [13]

Before (including deletions): the same model

After: Claude Opus 4.8

Change 576: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: The Opus 4.8 connectivity experiment is scored only against the human- verified nets of Section V-C.

Change 577: replace | old pp. [9] | new pp. [13]

Before (including deletions): evaluation scripts,

After: the

Change 578: replace | old pp. [9] | new pp. [13]

Before (including deletions): genera- tor,

After: generator,

Change 579: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: CGHD-derived reference nets of the held-out bench- mark (ground_truth/cghd_ref/), the audited nets of 37 held-out images, the verification timing log and the model pre-screen record (real_nets_working.json, timing_log.jsonl and model_check.json in ground_truth/cghd_ref_audit/), the

Change 580: delete | old pp. [9] | new pp. []

Before (including deletions): connected-component

After: (none)

Change 581: delete | old pp. [9] | new pp. []

Before (including deletions): and

After: (none)

Change 582: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: and the evaluation scripts for the paired tests, strata, rescaling, held-out and end-to-end experiments (wire_detection/benchmark/revision2/)

Change 583: replace | old pp. [9] | new pp. [13]

Before (including deletions): <https://github.com/boscochanam/circuit-digitization>; an

After: <https://github.com/boscochanam/circuit-digitization>, together with the image-to-drafter mapping used in Table 11. An

Change 584: replace | old pp. [9] | new pp. [13]

Before (including deletions): release

After: release, which predates the revision experiments,

Change 585: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: The released overlay images and component labels are derived from CGHD-1152 and remain under CC BY 4.0 [15]. The held-out reference nets and audit labels, derived from the CGHD v12 annotations [30], are released under CC BY-SA 4.0. The code and our own annotations are released under the MIT license (see `ground_truth/LICENSE`).

Change 586: replace | old pp. [9] | new pp. [13]

Before (including deletions): the CGHD-1152 dataset [1], [2]. The evaluation

After: CGHD

Change 587: replace | old pp. [9] | new pp. [13]

Before (including deletions): are drawn from CGHD-1152, which is distributed under a CC BY 4.0 licence [2]; our released overlays are adaptations of those images (grayscale conversion, upscaling, and ground-truth wire polylines drawn on top) and are redistributed under the same licence, while our net annotations are original work released under the MIT licence. VOLUME 11, 2023 9

After: [14], [15].

Change 588: delete | old pp. [10] | new pp. []

Before (including deletions): F. Thoma, J. Bayer, Y. Li, and A. Dengel, "A public ground-truth dataset for handwritten circuit diagram images," in Proc. Int. Conf. Document Analysis and Recognition (ICDAR), 2021, pp. 20–27. [Online]. Available: https://doi.org/10.1007/978-3-030-86198-8_2 [2] F. Thoma, J. Bayer, and Y. Li, "CGHD1152," Zenodo, 2022, dataset, licensed CC BY 4.0. [Online]. Available: <https://doi.org/10.5281/zenodo.6385814> [3] C. R. Kelly and J. M. Cole, "Digitizing images of electrical-circuit schematics," APL Mach. Learn., vol. 2, no. 1, p. 016109, 2024. [4]

After: (none)

Change 589: replace | old pp. [10] | new pp. [13]

Before (including deletions): [5] J. Bhandari, V. Bhat, Y. He, H. Rahmani, S. Garg, and R. Karri, "Masala- CHAI: A large-scale SPICE netlist dataset for analog circuits by harnessing AI," arXiv:2411.14299, 2024. [6]

After: [2]

Change 590: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: [3] J. Bhandari, V. Bhat, Y. He, H. Rahmani, S. Garg, and R. Karri, "Masala- CHAI: A large-scale SPICE netlist dataset for analog circuits by harnessing AI," arXiv:2411.14299, 2024. [4] C. R. Kelly and J. M. Cole, "Digitizing images of electrical-circuit schematics," APL Mach. Learn., vol. 2, no. 1, p. 016109, 2024. [5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 779–788. [6] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO," 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>

Change 591: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: VOLUME 11, 2023 13

Change 592: delete | old pp. [10] | new pp. []

Before (including deletions): J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 779–788. [11] G. Jocher,

A. Chaurasia, and J. Qiu, "Ultralytics YOLO," 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics> [12]

After: (none)

Change 593: replace | old pp. [10] | new pp. [14]

Before (including deletions): [13]

After: [11]

Change 594: insert | old pp. [] | new pp. [14]

Before (including deletions): (none)

After: [12] S. Gao, F. Shi, X. Xu, J. Chen, H. H.-C. Lu, et al., "A memristor- based neural network circuit with classical conditioning and fear generalization," IEEE Trans. Consumer Electron., early access, 2026, doi: 10.1109/TCE.2026.3663359. [13] S. Gao, Y. Song, Y. Cao, H. H.-C. Lu, L. Qin, Y. Zhang, and J. Mou, "Biologically plausible memristive decision-making circuit for adaptive control in industrial autonomous navigation," IEEE Trans. Ind. Informat., vol. 22, pp. 4801–4812, 2026, doi: 10.1109/TII.2026.3668794.

Change 595: insert | old pp. [] | new pp. [14]

Before (including deletions): (none)

After: F. Thoma, J. Bayer, Y. Li, and A. Dengel, "A public ground-truth dataset for handwritten circuit diagram images," in Proc. Int. Conf. Document Analysis and Recognition (ICDAR), 2021, pp. 20–27. [Online]. Available: https://doi.org/10.1007/978-3-030-86198-8_2 [15] F. Thoma, J. Bayer, and Y. Li, "CGHD1152," Zenodo, 2021, dataset, licensed CC BY 4.0. [Online]. Available: <https://doi.org/10.5281/zenodo.6385814> [16]

Change 596: replace | old pp. [10] | new pp. [14]

Before (including deletions): [15]

After: [17]

Change 597: replace | old pp. [10] | new pp. [14]

Before (including deletions): [16]

After: [18]

Change 598: replace | old pp. [10] | new pp. [14]

Before (including deletions): [17]

After: [19]

Change 599: replace | old pp. [10] | new pp. [14]

Before (including deletions): [18]

After: [20]

Change 600: replace | old pp. [10] | new pp. [14]

Before (including deletions): Kulkarni

After: Kulkarni, V. Kumar, R. L. Minz, M. Varshney, T. Samon, A. Mitra,

Change 601: replace | old pp. [10] | new pp. [14]

Before (including deletions): ICCV Workshops,

After: IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW),

Change 602: replace | old pp. [10] | new pp. [14]

Before (including deletions): [19]

After: [21]

Change 603: replace | old pp. [10] | new pp. [14]

Before (including deletions): [20]

After: [22]

Change 604: replace | old pp. [10] | new pp. [14]

Before (including deletions): [21]

After: [23]

Change 605: replace | old pp. [10] | new pp. [14]

Before (including deletions): [22]

After: [24]

Change 606: replace | old pp. [10] | new pp. [14]

Before (including deletions): [23]

After: [25]

Change 607: replace | old pp. [10] | new pp. [14]

Before (including deletions): [24]

After: [26]

Change 608: replace | old pp. [10] | new pp. [14]

Before (including deletions): [25]

After: [27]

Change 609: replace | old pp. [10] | new pp. [14]

Before (including deletions): [26]

After: [28]

Change 610: replace | old pp. [10] | new pp. [14]

Before (including deletions): [27]

After: [29]

Change 611: insert | old pp. [] | new pp. [14]

Before (including deletions): (none)

After: [30] F. Thoma, J. Bayer, Y. Li, and A. Dengel, "A public ground-truth dataset for handwritten circuit diagram images," Zenodo, 2023, dataset, version 12. [Online]. Available: <https://doi.org/10.5281/zenodo.10056817>

Change 612: delete | old pp. [10] | new pp. []

Before (including deletions): 10 VOLUME 11, 2023

After: (none)

Change 613: insert | old pp. [] | new pp. [14]

Before (including deletions): (none)

After: 14 VOLUME 11, 2023

Change 614: replace | old pp. [11] | new pp. [15]

Before (including deletions): 11

After: 15